

Combined Modern Linear Algebra Conspect

Vector spaces, bases, coordinates, operators, canonical forms, spectral theory,
bilinear and quadratic forms

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Convention 0.1 (Global assumptions). Unless stated otherwise, $\mathbb{K}$ is a field and all vector spaces are finite-dimensional over $\mathbb{K}$. For inner product spaces, $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. In the complex case the inner product is taken to be linear in the first argument and conjugate-linear in the second argument:

$$\langle \alpha x_1 + \beta x_2, y \rangle = \alpha \langle x_1, y \rangle + \beta \langle x_2, y \rangle, \quad \langle x, \alpha y_1 + \beta y_2 \rangle = \bar{\alpha} \langle x, y_1 \rangle + \bar{\beta} \langle x, y_2 \rangle.$$

The reason for this convention is that a functional represented by a vector y has the form $x \mapsto \langle x, y \rangle$.

1. Vector spaces: definition and examples

Core idea. A vector space is a set whose elements can be added and scaled, with exactly the algebraic rules needed for linear combinations.

Definition 1.1 (Vector space). A *vector space over $\mathbb{K}$* is a set V equipped with two operations

$$+ : V \times V \rightarrow V, \quad \mathbb{K} \times V \rightarrow V,$$

called addition and scalar multiplication, such that for all $u, v, w \in V$ and $a, b \in \mathbb{K}$:

- (i) $u + v = v + u$;
- (ii) $(u + v) + w = u + (v + w)$;
- (iii) there exists $0 \in V$ such that $v + 0 = v$;
- (iv) for every v there exists $-v$ such that $v + (-v) = 0$;
- (v) $a(u + v) = au + av$;
- (vi) $(a + b)v = av + bv$;
- (vii) $(ab)v = a(bv)$;
- (viii) $1v = v$.

The elements of V are called *vectors*; the elements of $\mathbb{K}$ are called *scalars*.

Proposition 1.2 (Basic consequences). *For every vector space V over $\mathbb{K}$ and every $v \in V$, $0_{\mathbb{K}}v = 0_V$, $a0_V = 0_V$, and $(-1)v = -v$.*

Proof. First,

$$0_{\mathbb{K}}v = (0_{\mathbb{K}} + 0_{\mathbb{K}})v = 0_{\mathbb{K}}v + 0_{\mathbb{K}}v.$$

Add the additive inverse of $0_{\mathbb{K}}v$ to both sides; this gives $0_V = 0_{\mathbb{K}}v$. Next,

$$a0_V = a(0_V + 0_V) = a0_V + a0_V,$$

so $a0_V = 0_V$. Finally,

$$v + (-1)v = 1v + (-1)v = (1 - 1)v = 0_{\mathbb{K}}v = 0_V,$$

so $(-1)v$ is the additive inverse of v , hence $(-1)v = -v$ by uniqueness of additive inverse. $\square$

Example 1.3 (Coordinate spaces). $\mathbb{K}^n$ is a vector space with componentwise operations:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n),$$

$$a(x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The vector space axioms follow component by component from the field axioms in $\mathbb{K}$.

Verification. For example, commutativity of addition follows because

$$(x_i) + (y_i) = (y_i) + (x_i)$$

in every coordinate. The distributive law follows because

$$a(x_i + y_i) = ax_i + ay_i$$

in every coordinate. The other axioms are identical componentwise reductions to the field axioms. $\square$

Example 1.4 (Polynomial spaces). $\mathbb{K}[x]$, the set of all polynomials with coefficients in $\mathbb{K}$, is a vector space under usual addition and scalar multiplication. For each $n \geq 0$, the subset

$$\mathbb{K}_{\leq n}[x] = \{p \in \mathbb{K}[x] : \deg p \leq n\} \cup \{0\}$$

is also a vector space.

Verification. The sum of two polynomials is a polynomial, and a scalar multiple of a polynomial is a polynomial. The axioms follow coefficientwise from the axioms of $\mathbb{K}$. If $p, q \in \mathbb{K}_{\leq n}[x]$, then $\deg(p+q) \leq n$ and $\deg(ap) \leq n$ unless the result is zero; hence $\mathbb{K}_{\leq n}[x]$ is closed under the two operations, and therefore inherits the vector space axioms from $\mathbb{K}[x]$. $\square$

Example 1.5 (Function spaces). For any set X , the set $\mathbb{K}^X$ of all functions $f : X \rightarrow \mathbb{K}$ is a vector space with pointwise operations:

$$(f + g)(x) = f(x) + g(x), \quad (af)(x) = af(x).$$

Verification. Closure is clear because $f(x) + g(x) \in \mathbb{K}$ and $af(x) \in \mathbb{K}$. Each vector space axiom holds after evaluating at an arbitrary point $x \in X$ and using the corresponding field axiom in $\mathbb{K}$. Since two functions are equal if and only if they agree at every point, the axioms hold for functions. $\square$

Definition 1.6 (Subspace). A subset $U \subseteq V$ is a *linear subspace* if $0 \in U$ and for all $u_1, u_2 \in U$, $a, b \in \mathbb{K}$,

$$au_1 + bu_2 \in U.$$

Proposition 1.7 (Subspace criterion). A nonempty subset $U \subseteq V$ is a subspace if and only if it is closed under all linear combinations $au + bv$.

Proof. If U is a subspace, then closure under $au + bv$ is part of the definition. Conversely, assume U is nonempty and closed under $au + bv$. Choose $u \in U$. With $a = b = 0$ and $v = u$, we get $0u + 0u = 0 \in U$. With arbitrary $u, v \in U$ and arbitrary $a, b \in \mathbb{K}$, closure gives $au + bv \in U$. Thus U is a vector space under inherited operations. $\square$

2. Equal cardinality of bases and transition matrices

Core idea. Dimension is intrinsic: bases are coordinate systems, but their number of vectors is independent of the chosen coordinates.

Definition 2.1 (Linear combination, span, independence, basis). Let $S \subseteq V$. A *linear combination* of vectors $s_1, \dots, s_m \in S$ is a vector

$$a_1 s_1 + \dots + a_m s_m, \quad a_i \in \mathbb{K}.$$

The *span* of S is

$$\text{span } S = \{a_1 s_1 + \dots + a_m s_m : m \geq 0, s_i \in S, a_i \in \mathbb{K}\}.$$

A finite list $(v_1, \dots, v_m)$ is *linearly independent* if

$$a_1 v_1 + \dots + a_m v_m = 0 \implies a_1 = \dots = a_m = 0.$$

It is *spanning* if $\text{span}(v_1, \dots, v_m) = V$. A *basis* is a linearly independent spanning list.

Lemma 2.2 (Exchange lemma). Let $B = (b_1, \dots, b_n)$ be a basis of V , and let $v \neq 0$. If

$$v = \alpha_1 b_1 + \dots + \alpha_n b_n$$

with $\alpha_j \neq 0$, then replacing b_j by v gives another basis:

$$(b_1, \dots, b_{j-1}, v, b_{j+1}, \dots, b_n).$$

Proof. Since $\alpha_j \neq 0$, solve the displayed formula for b_j :

$$b_j = \alpha_j^{-1} v - \sum_{i \neq j} \alpha_j^{-1} \alpha_i b_i.$$

Thus the old vector b_j lies in the span of the new list. Every old basis vector is therefore in the span of the new list, so the new list spans V .

The new list has n vectors and spans V . To prove it is independent, suppose

$$\beta v + \sum_{i \neq j} \beta_i b_i = 0.$$

Substitute $v = \sum_i \alpha_i b_i$:

$$(\beta \alpha_j) b_j + \sum_{i \neq j} (\beta \alpha_i + \beta_i) b_i = 0.$$

Because B is independent, $\beta \alpha_j = 0$. Since $\alpha_j \neq 0$, $\beta = 0$, and then every $\beta_i = 0$. Hence the new list is independent. $\square$

Lemma 2.3 (Steinitz replacement principle). Let $B = (b_1, \dots, b_n)$ be a basis of V , and let $S = (s_1, \dots, s_m)$ be linearly independent. Then $m \leq n$. More precisely, after replacing m suitable vectors of B by $s_1, \dots, s_m$, one still has a basis.

Proof. We prove the stronger statement by induction on m .

For $m = 0$ there is nothing to prove. Suppose the result is true for $m - 1$. Then, after a reordering of the current basis, there is a basis

$$(s_1, \dots, s_{m-1}, c_m, \dots, c_n).$$

Since this list spans V , write

$$s_m = \alpha_1 s_1 + \dots + \alpha_{m-1} s_{m-1} + \alpha_m c_m + \dots + \alpha_n c_n.$$

At least one coefficient among $\alpha_m, \dots, \alpha_n$ is nonzero; otherwise $s_m \in \text{span}(s_1, \dots, s_{m-1})$, contradicting independence of S . Choose $j \geq m$ with $\alpha_j \neq 0$. By the exchange lemma, replacing c_j by s_m gives a basis. Hence the statement holds for m . In particular, the induction cannot continue past n , so $m \leq n$. $\square$

Theorem 2.4 (Equal cardinality of bases). *Any two finite bases of a vector space V have the same number of vectors.*

Proof. Let B and C be bases, with $|B| = n$ and $|C| = m$. Since C is linearly independent and B is a basis, Steinitz gives $m \leq n$. Since B is linearly independent and C is a basis, Steinitz gives $n \leq m$. Hence $m = n$. $\square$

Definition 2.5 (Dimension). If V has a finite basis, the common number of vectors in any basis is called the *dimension* of V and is denoted $\dim V$.

Definition 2.6 (Transition matrix). Let $E = (e_1, \dots, e_n)$ and $F = (f_1, \dots, f_n)$ be ordered bases of V . The *transition matrix from F -coordinates to E -coordinates* is

$$P_{E \leftarrow F} = \begin{bmatrix} [f_1]_E & [f_2]_E & \cdots & [f_n]_E \end{bmatrix}.$$

Equivalently, its j th column is the coordinate column of f_j in the basis E .

Proposition 2.7 (Transition matrix formula). *For every $v \in V$,*

$$[v]_E = P_{E \leftarrow F} [v]_F.$$

Moreover, $P_{E \leftarrow F}$ is invertible and

$$(P_{E \leftarrow F})^{-1} = P_{F \leftarrow E}.$$

Proof. Write $[v]_F = (a_1, \dots, a_n)^T$, so $v = \sum_j a_j f_j$. Each f_j has E -coordinate column $[f_j]_E$. Therefore

$$[v]_E = \sum_{j=1}^n a_j [f_j]_E = P_{E \leftarrow F} [v]_F.$$

Applying the same formula with E and F exchanged gives $[v]_F = P_{F \leftarrow E} [v]_E$. Hence

$$[v]_E = P_{E \leftarrow F} P_{F \leftarrow E} [v]_E$$

for all coordinate columns $[v]_E \in \mathbb{K}^n$. Thus $P_{E \leftarrow F} P_{F \leftarrow E} = I$. Similarly $P_{F \leftarrow E} P_{E \leftarrow F} = I$. Therefore the matrices are inverse. $\square$

3. Coordinates of a vector and change of basis

Core idea. Coordinates are not the vector itself; they are the vector's coefficients relative to a chosen basis.

Definition 3.1 (Coordinate map). Let $E = (e_1, \dots, e_n)$ be a basis of V . The *coordinate map* associated to E is

$$\Phi_E : V \rightarrow \mathbb{K}^n, \quad \Phi_E(v) = [v]_E,$$

where $[v]_E = (a_1, \dots, a_n)^T$ means

$$v = a_1 e_1 + \cdots + a_n e_n.$$

Theorem 3.2 (Existence and uniqueness of coordinates). *For every basis $E = (e_1, \dots, e_n)$ and every $v \in V$, there is a unique coordinate column $[v]_E \in \mathbb{K}^n$ such that*

$$v = a_1 e_1 + \cdots + a_n e_n.$$

Proof. Existence follows from the fact that E spans V . Thus $v = \sum_i a_i e_i$ for some scalars a_i .

For uniqueness, suppose

$$v = \sum_i a_i e_i = \sum_i b_i e_i.$$

Subtracting gives

$$0 = \sum_i (a_i - b_i) e_i.$$

Since E is linearly independent, $a_i - b_i = 0$ for every i . Hence $a_i = b_i$ for every i . $\square$

Proposition 3.3 (Coordinate map is an isomorphism). *The coordinate map $\Phi_E : V \rightarrow \mathbb{K}^n$ is a linear isomorphism.*

Proof. Let $v = \sum_i a_i e_i$ and $w = \sum_i b_i e_i$. Then

$$v + w = \sum_i (a_i + b_i) e_i,$$

so $[v + w]_E = [v]_E + [w]_E$. Also, for $\lambda \in \mathbb{K}$,

$$\lambda v = \sum_i (\lambda a_i) e_i,$$

so $[\lambda v]_E = \lambda [v]_E$. Thus Φ_E is linear.

It is injective because $[v]_E = 0$ implies $v = 0$. It is surjective because for every $(a_1, \dots, a_n)^T \in \mathbb{K}^n$, the vector $\sum_i a_i e_i$ has precisely that coordinate column. Therefore Φ_E is a linear isomorphism. $\square$

Theorem 3.4 (Change of coordinates). *Let E and F be ordered bases of V . Then*

$$[v]_E = P_{E \leftarrow F} [v]_F, \quad [v]_F = P_{F \leftarrow E} [v]_E = (P_{E \leftarrow F})^{-1} [v]_E.$$

Proof. The first equality is exactly the transition matrix formula. The second follows by applying the first equality with E and F interchanged and by the inverse identity $P_{F \leftarrow E} = (P_{E \leftarrow F})^{-1}$. $\square$

Remark 3.5 (Passive versus active interpretation). Changing coordinates from E to F does not move the vector v . It changes the name of the same vector. In contrast, applying a linear operator to v changes the vector itself. This is why change-of-basis matrices and operator matrices have different conceptual roles, although both are matrices.

4. Coordinates in an orthogonal basis and Fourier coefficients

Core idea. Orthogonality turns coordinate extraction into taking inner products.

Definition 4.1 (Inner product space). An *inner product space* over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is a vector space V with a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{K}$$

such that for all $x, y, z \in V$ and $a, b \in \mathbb{K}$:

- (i) $\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$;
- (ii) $\langle x, y \rangle = \overline{\langle y, x \rangle}$;
- (iii) $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

The induced norm is $\|x\| = \sqrt{\langle x, x \rangle}$.

Definition 4.2 (Orthogonal and orthonormal bases). A basis $(e_1, \dots, e_n)$ is *orthogonal* if $\langle e_i, e_j \rangle = 0$ for $i \neq j$. It is *orthonormal* if, in addition, $\|e_i\| = 1$ for all i .

Theorem 4.3 (Coordinates in an orthogonal basis). Let $E = (e_1, \dots, e_n)$ be an orthogonal basis of an inner product space V . Then every $v \in V$ has expansion

$$v = \sum_{i=1}^n \frac{\langle v, e_i \rangle}{\langle e_i, e_i \rangle} e_i.$$

If E is orthonormal, then

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

The coefficients

$$\hat{v}_i = \frac{\langle v, e_i \rangle}{\langle e_i, e_i \rangle}$$

are called the *finite Fourier coefficients* of v in the orthogonal basis E .

Proof. Write $v = \sum_i a_i e_i$. Take the inner product with e_j :

$$\langle v, e_j \rangle = \left\langle \sum_i a_i e_i, e_j \right\rangle = \sum_i a_i \langle e_i, e_j \rangle = a_j \langle e_j, e_j \rangle.$$

Since $e_j \neq 0$, positive-definiteness gives $\langle e_j, e_j \rangle > 0$. Therefore

$$a_j = \frac{\langle v, e_j \rangle}{\langle e_j, e_j \rangle}.$$

If E is orthonormal, $\langle e_j, e_j \rangle = 1$, giving $a_j = \langle v, e_j \rangle$. □

Proposition 4.4 (Finite Parseval identity). If $E = (e_1, \dots, e_n)$ is an orthonormal basis of V , then

$$\|v\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2.$$

Proof. Using the orthonormal expansion $v = \sum_i a_i e_i$ with $a_i = \langle v, e_i \rangle$,

$$\|v\|^2 = \left\langle \sum_i a_i e_i, \sum_j a_j e_j \right\rangle = \sum_{i,j} a_i \overline{a_j} \langle e_i, e_j \rangle = \sum_i a_i \overline{a_i} = \sum_i |a_i|^2.$$

Substituting $a_i = \langle v, e_i \rangle$ gives the result. □

5. Projection onto a subspace and the extremal property

Core idea. Orthogonal projection is the unique nearest point in a subspace.

Definition 5.1 (Orthogonal complement). Let U be a subspace of an inner product space V . Its *orthogonal complement* is

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for every } u \in U\}.$$

Proposition 5.2 ($U^\perp$ is a subspace). *For every subspace $U \subseteq V$, $U^\perp$ is a subspace of V .*

Proof. Clearly $0 \in U^\perp$. If $x, y \in U^\perp$ and $a, b \in \mathbb{K}$, then for every $u \in U$,

$$\langle ax + by, u \rangle = a \langle x, u \rangle + b \langle y, u \rangle = 0.$$

Thus $ax + by \in U^\perp$. □

Lemma 5.3 (Existence of orthonormal bases in finite-dimensional subspaces). *Every finite-dimensional inner product space has an orthonormal basis.*

Proof. Take any basis and apply the Gram-Schmidt construction proved in Section 6. The construction produces an orthogonal basis spanning the same space; normalizing its nonzero vectors gives an orthonormal basis. □

Theorem 5.4 (Orthogonal decomposition). *Let U be a finite-dimensional subspace of an inner product space V . Then every $v \in V$ has a unique decomposition*

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

Equivalently,

$$V = U \oplus U^\perp.$$

Proof. Choose an orthonormal basis $(e_1, \dots, e_m)$ of U . Define

$$u = \sum_{i=1}^m \langle v, e_i \rangle e_i, \quad w = v - u.$$

Then $u \in U$. For each j ,

$$\langle w, e_j \rangle = \langle v, e_j \rangle - \sum_i \langle v, e_i \rangle \langle e_i, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0.$$

Since every element of U is a linear combination of the e_j , it follows that $w \in U^\perp$. Thus existence is proved.

For uniqueness, suppose $v = u_1 + w_1 = u_2 + w_2$ with $u_i \in U$ and $w_i \in U^\perp$. Then

$$u_1 - u_2 = w_2 - w_1.$$

The left side lies in U and the right side lies in $U^\perp$. Hence this vector lies in $U \cap U^\perp$. If $z \in U \cap U^\perp$, then $\langle z, z \rangle = 0$, so $z = 0$. Therefore $u_1 = u_2$ and $w_1 = w_2$. □

Definition 5.5 (Orthogonal projection). The map

$$P_U : V \rightarrow U, \quad P_U(v) = u$$

where $v = u + w$ is the orthogonal decomposition, is called the *orthogonal projection onto U* .

Proposition 5.6 (Formula for projection). *If $(e_1, \dots, e_m)$ is an orthonormal basis of U , then*

$$P_U v = \sum_{i=1}^m \langle v, e_i \rangle e_i.$$

If $(u_1, \dots, u_m)$ is merely orthogonal, then

$$P_U v = \sum_{i=1}^m \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

Proof. The proof of the orthogonal decomposition already showed the orthonormal formula. For an orthogonal basis (u_i) , the normalized vectors $e_i = u_i / \|u_i\|$ form an orthonormal basis. Substituting e_i into the orthonormal formula gives

$$\langle v, u_i / \|u_i\| \rangle \frac{u_i}{\|u_i\|} = \frac{\langle v, u_i \rangle}{\|u_i\|^2} u_i = \frac{\langle v, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

□

Theorem 5.7 (Extremal property of projection). *Let U be a finite-dimensional subspace of V . For every $v \in V$, $P_U v$ is the unique point of U nearest to v :*

$$\|v - P_U v\| \leq \|v - u\| \quad \text{for all } u \in U,$$

with equality if and only if $u = P_U v$.

Proof. Write $v = P_U v + w$ with $w \in U^\perp$. For any $u \in U$,

$$v - u = w + (P_U v - u).$$

The two summands are orthogonal because $w \in U^\perp$ and $P_U v - u \in U$. By Pythagoras,

$$\|v - u\|^2 = \|w\|^2 + \|P_U v - u\|^2 \geq \|w\|^2 = \|v - P_U v\|^2.$$

Equality holds if and only if $\|P_U v - u\|^2 = 0$, hence if and only if $u = P_U v$. □

Corollary 5.8 (Distance to a subspace).

$$\text{dist}(v, U) = \inf_{u \in U} \|v - u\| = \|v - P_U v\|.$$

Proof. The extremal property shows that $\|v - P_U v\|$ is a lower bound for all distances $\|v - u\|$ and is attained at $u = P_U v$. □

6. Gram-Schmidt orthogonalization

Core idea. Gram-Schmidt subtracts from each new vector its already constructed projections, leaving the new orthogonal component.

Construction 6.1 (Gram-Schmidt). Let $(v_1, \dots, v_m)$ be a linearly independent list in an inner product space. Define recursively

$$\begin{aligned} w_1 &= v_1, \\ w_k &= v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle} w_i, \quad k = 2, \dots, m. \end{aligned}$$

Then define

$$e_k = \frac{w_k}{\|w_k\|}.$$

Theorem 6.2 (Gram-Schmidt theorem). *The vectors $(w_1, \dots, w_m)$ are nonzero and orthogonal. The vectors $(e_1, \dots, e_m)$ are orthonormal. Moreover, for each k ,*

$$\text{span}(v_1, \dots, v_k) = \text{span}(w_1, \dots, w_k) = \text{span}(e_1, \dots, e_k).$$

Proof. We prove the statement by induction on k .

For $k = 1$, $w_1 = v_1 \neq 0$ because the list is linearly independent. The span equality is immediate.

Assume the statement is true up to $k - 1$. The formula for w_k expresses w_k as v_k plus a vector in $\text{span}(w_1, \dots, w_{k-1}) = \text{span}(v_1, \dots, v_{k-1})$. Therefore

$$\text{span}(w_1, \dots, w_k) = \text{span}(v_1, \dots, v_k)$$

provided $w_k \neq 0$.

If $w_k = 0$, then the defining formula gives

$$v_k \in \text{span}(w_1, \dots, w_{k-1}) = \text{span}(v_1, \dots, v_{k-1}),$$

contradicting linear independence of $(v_1, \dots, v_k)$. Hence $w_k \neq 0$.

For $j < k$,

$$\langle w_k, w_j \rangle = \langle v_k, w_j \rangle - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle} \langle w_i, w_j \rangle.$$

By the induction hypothesis, $\langle w_i, w_j \rangle = 0$ for $i \neq j$, and $\langle w_j, w_j \rangle$ remains when $i = j$. Thus

$$\langle w_k, w_j \rangle = \langle v_k, w_j \rangle - \frac{\langle v_k, w_j \rangle}{\langle w_j, w_j \rangle} \langle w_j, w_j \rangle = 0.$$

So the new vector is orthogonal to the previous ones.

Normalizing nonzero orthogonal vectors gives an orthonormal list, and normalization does not change spans. This completes the induction. $\square$

Corollary 6.3 (Existence of orthonormal bases). *Every finite-dimensional inner product space has an orthonormal basis.*

Proof. Take any basis and apply Gram-Schmidt. The resulting orthonormal list spans the same space, hence is an orthonormal basis. $\square$

7. Linear operators: definition, properties, and operations

Core idea. A linear operator is a structure-preserving map: it preserves all finite linear combinations.

Definition 7.1 (Linear map and linear operator). Let V, W be vector spaces over the same field $\mathbb{K}$. A map $A : V \rightarrow W$ is *linear* if

$$A(\alpha x + \beta y) = \alpha Ax + \beta Ay$$

for all $x, y \in V$ and all $\alpha, \beta \in \mathbb{K}$. If $V = W$, a linear map $A : V \rightarrow V$ is called a *linear operator on* V . The set of all linear maps $V \rightarrow W$ is denoted $\text{Hom}(V, W)$; the set of all operators on V is denoted $\text{End}(V)$.

Proposition 7.2 (Elementary properties). *If $A : V \rightarrow W$ is linear, then:*

- (i) $A0 = 0$;
- (ii) $A(-v) = -Av$;
- (iii) for every finite linear combination,

$$A\left(\sum_{i=1}^m a_i v_i\right) = \sum_{i=1}^m a_i Av_i.$$

Proof. (i) $A0 = A(0 + 0) = A0 + A0$, so subtracting $A0$ gives $A0 = 0$.

(ii) $0 = A0 = A(v + (-v)) = Av + A(-v)$, hence $A(-v) = -Av$.

(iii) The formula for two terms is exactly linearity. The general finite formula follows by induction on m . $\square$

Proposition 7.3 ($\text{Hom}(V, W)$ is a vector space). *Define*

$$(A + B)(v) = Av + Bv, \quad (\lambda A)(v) = \lambda Av.$$

Then $\text{Hom}(V, W)$ is a vector space over $\mathbb{K}$.

Proof. First, $A + B$ is linear because

$$(A + B)(\alpha x + \beta y) = A(\alpha x + \beta y) + B(\alpha x + \beta y) = \alpha(Ax + Bx) + \beta(Ay + By).$$

Similarly, λA is linear. The vector space axioms hold pointwise: for example, $(A + B)(v) = A(v) + B(v)$, and addition in W is associative and commutative. The zero element is the zero map $0(v) = 0_W$, and the additive inverse is $(-A)(v) = -A(v)$. $\square$

Proposition 7.4 (Composition). *If $A : U \rightarrow V$ and $B : V \rightarrow W$ are linear, then $B \circ A : U \rightarrow W$ is linear. Composition is associative, and identity maps act as identities:*

$$C \circ (B \circ A) = (C \circ B) \circ A, \quad A \circ \text{id}_U = A, \quad \text{id}_V \circ A = A.$$

Proof. For linearity,

$$(B \circ A)(\alpha x + \beta y) = B(\alpha Ax + \beta Ay) = \alpha BAx + \beta BAy.$$

Associativity and the identity laws follow from the corresponding elementary facts about composition of functions. $\square$

Definition 7.5 (Kernel and image). For a linear map $A : V \rightarrow W$,

$$\text{Ker } A = \{v \in V : Av = 0\}, \quad \text{Im } A = \{Av : v \in V\}.$$

Proposition 7.6 (Kernel and image are subspaces). *$\text{Ker } A$ is a subspace of V , and $\text{Im } A$ is a subspace of W .*

Proof. If $x, y \in \text{Ker } A$, then

$$A(\alpha x + \beta y) = \alpha Ax + \beta Ay = 0,$$

so $\alpha x + \beta y \in \text{Ker } A$. Hence $\text{Ker } A$ is a subspace.

If $u = Ax$ and $v = Ay$ lie in $\text{Im } A$, then

$$\alpha u + \beta v = \alpha Ax + \beta Ay = A(\alpha x + \beta y),$$

so $\alpha u + \beta v \in \text{Im } A$. Hence $\text{Im } A$ is a subspace. $\square$

Theorem 7.7 (Rank-nullity theorem). *If $A : V \rightarrow W$ is linear and V is finite-dimensional, then*

$$\dim V = \dim \text{Ker } A + \dim \text{Im } A.$$

Proof. Let $(k_1, \dots, k_r)$ be a basis of $\text{Ker } A$. Extend it to a basis of V :

$$(k_1, \dots, k_r, v_1, \dots, v_s).$$

We claim that $(Av_1, \dots, Av_s)$ is a basis of $\text{Im } A$.

Spanning: for any $x \in V$,

$$x = \sum_i a_i k_i + \sum_j b_j v_j,$$

so

$$Ax = \sum_j b_j Av_j.$$

Thus the Av_j span $\text{Im } A$.

Independence: if $\sum_j b_j Av_j = 0$, then

$$A \left(\sum_j b_j v_j \right) = 0,$$

so $\sum_j b_j v_j \in \text{Ker } A$. Hence it can be written as $\sum_i a_i k_i$. Therefore

$$\sum_j b_j v_j - \sum_i a_i k_i = 0.$$

Since the extended list is a basis, all coefficients vanish, in particular all $b_j = 0$. Thus the Av_j are independent. Therefore $\dim \text{Im } A = s$, and $\dim V = r + s$. $\square$

8. Matrices of linear operators and change of bases

Core idea. A matrix is not the operator itself; it is the coordinate expression of the operator after choosing bases.

Definition 8.1 (Matrix of a linear map). Let $A : V \rightarrow W$ be linear, let $E = (e_1, \dots, e_n)$ be a basis of V , and let $F = (f_1, \dots, f_m)$ be a basis of W . The matrix of A from E to F is

$$[A]_{F \leftarrow E} = \begin{bmatrix} [Ae_1]_F & [Ae_2]_F & \cdots & [Ae_n]_F \end{bmatrix} \in \text{Mat}_{m \times n}(\mathbb{K}).$$

If $A : V \rightarrow V$ and the same basis E is used in domain and codomain, we write $[A]_E = [A]_{E \leftarrow E}$.

Theorem 8.2 (Coordinate formula for a matrix). *For every $v \in V$,*

$$[Av]_F = [A]_{F \leftarrow E} [v]_E.$$

Proof. Write $[v]_E = (a_1, \dots, a_n)^T$, so $v = \sum_j a_j e_j$. By linearity,

$$Av = \sum_j a_j Ae_j.$$

Taking F -coordinates gives

$$[Av]_F = \sum_j a_j [Ae_j]_F.$$

This is precisely the product of the matrix whose j th column is $[Ae_j]_F$ with the column (a_j) . $\square$

Theorem 8.3 (Change of matrix under change of bases). *Let $A : V \rightarrow W$ be linear. Let E, E' be bases of V and F, F' bases of W . Put*

$$M = [A]_{F \leftarrow E}, \quad M' = [A]_{F' \leftarrow E'},$$

$$P = P_{E \leftarrow E'}, \quad Q = P_{F \leftarrow F'}.$$

Then

$$M' = Q^{-1}MP.$$

For an operator $A : V \rightarrow V$, if E, E' are bases and $P = P_{E \leftarrow E'}$, then

$$[A]_{E'} = P^{-1}[A]_E P.$$

Proof. For every $v \in V$,

$$[v]_E = P[v]_{E'}.$$

Also, because $Q = P_{F \leftarrow F'}$, we have

$$[Av]_F = Q[Av]_{F'}, \quad \text{hence} \quad [Av]_{F'} = Q^{-1}[Av]_F.$$

Using the old matrix M ,

$$[Av]_{F'} = Q^{-1}[Av]_F = Q^{-1}M[v]_E = Q^{-1}MP[v]_{E'}.$$

Since this holds for every coordinate column $[v]_{E'}$, the matrix in the new bases is $M' = Q^{-1}MP$.

If $V = W$ and the same basis is used in domain and codomain, then $Q = P$, so $[A]_{E'} = P^{-1}[A]_E P$. $\square$

Definition 8.4 (Similar matrices). Two square matrices $M, N \in \text{Mat}_n(\mathbb{K})$ are *similar* if there exists $P \in \text{GL}_n(\mathbb{K})$ such that

$$N = P^{-1}MP.$$

Proposition 8.5 (Same operator, similar matrices). *Two matrices represent the same operator in two different bases if and only if they are similar.*

Proof. If $M = [A]_E$ and $N = [A]_{E'}$, the change-of-basis theorem gives $N = P^{-1}MP$ for $P = P_{E \leftarrow E'}$, so they are similar.

Conversely, if $N = P^{-1}MP$ and $M = [A]_E$, choose the basis E' whose transition matrix to E is P . That means the columns of P are the E -coordinates of the vectors of E' . Since P is invertible, those vectors form a basis. The change-of-basis formula gives $[A]_{E'} = N$. $\square$

9. Almost diagonal canonical form: Jordan form

Core idea. If diagonalization fails, generalized eigenvectors still decompose the operator into Jordan blocks: diagonal entries are eigenvalues and the only possible off-diagonal entries are ones inside chains.

Convention 9.1. This section assumes that $\mathbb{K}$ is algebraically closed, for example $\mathbb{K} = \mathbb{C}$. Over a non-algebraically closed field, not every characteristic polynomial splits, so Jordan form over $\mathbb{K}$ may fail to exist.

Definition 9.2 (Jordan block). For $\lambda \in \mathbb{K}$ and $r \geq 1$, the Jordan block of size r with eigenvalue λ is

$$J_r(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}.$$

A *Jordan matrix* is a block diagonal matrix whose blocks are Jordan blocks.

Definition 9.3 (Generalized eigenspace). For an operator $A \in \text{End}(V)$ and $\lambda \in \mathbb{K}$, the generalized eigenspace is

$$G_\lambda(A) = \bigcup_{r \geq 1} \text{Ker}(A - \lambda I)^r.$$

Since V is finite-dimensional, this union stabilizes: for all sufficiently large r ,

$$G_\lambda(A) = \text{Ker}(A - \lambda I)^r.$$

Lemma 9.4 (Stabilization of kernels). *For any operator T on a finite-dimensional space, the chain*

$$\text{Ker } T \subseteq \text{Ker } T^2 \subseteq \text{Ker } T^3 \subseteq \cdots$$

stabilizes. If $\text{Ker } T^r = \text{Ker } T^{r+1}$, then $\text{Ker } T^r = \text{Ker } T^{r+s}$ for all $s \geq 0$.

Proof. The dimensions of $\text{Ker } T^r$ form a nondecreasing sequence of integers bounded above by $\dim V$, so the sequence stabilizes. If $\text{Ker } T^r = \text{Ker } T^{r+1}$ and $x \in \text{Ker } T^{r+2}$, then $Tx \in \text{Ker } T^{r+1} = \text{Ker } T^r$, so $T^{r+1}x = 0$, hence $x \in \text{Ker } T^{r+1} = \text{Ker } T^r$. Thus $\text{Ker } T^{r+2} \subseteq \text{Ker } T^r$, and the reverse inclusion is automatic. Repeating proves the claim for all s . $\square$

Lemma 9.5 (Cayley-Hamilton theorem). *For every square matrix $M \in \text{Mat}_n(\mathbb{K})$,*

$$\chi_M(M) = 0.$$

Equivalently, for every operator $A \in \text{End}(V)$,

$$\chi_A(A) = 0.$$

Proof. We prove the matrix form. Work in the polynomial ring $\mathbb{K}[t]$. The adjugate identity gives

$$\text{adj}(tI - M)(tI - M) = \det(tI - M)I = \chi_M(t)I.$$

Write

$$\text{adj}(tI - M) = B_0 + B_1t + \cdots + B_{n-1}t^{n-1}$$

with matrix coefficients $B_i \in \text{Mat}_n(\mathbb{K})$, and write

$$\chi_M(t) = c_0 + c_1t + \cdots + c_{n-1}t^{n-1} + t^n.$$

Comparing coefficients in

$$(B_0 + B_1t + \cdots + B_{n-1}t^{n-1})(tI - M) = \chi_M(t)I$$

gives

$$-B_0M = c_0I,$$

$$B_{k-1} - B_k M = c_k I \quad (1 \leq k \leq n-1),$$

$$B_{n-1} = I.$$

Multiply these identities respectively by $I, M, M^2, \dots, M^n$ and add them:

$$\begin{aligned} -B_0 M + (B_0 M - B_1 M^2) + \dots + (B_{n-2} M^{n-1} - B_{n-1} M^n) + B_{n-1} M^n \\ = c_0 I + c_1 M + \dots + c_{n-1} M^{n-1} + M^n. \end{aligned}$$

The left side telescopes to 0, so the right side is $\chi_M(M)$. Hence $\chi_M(M) = 0$.

For an operator A , choose a basis and apply the matrix result to $[A]_E$. The coordinate matrix of $\chi_A(A)$ is $\chi_{[A]_E}([A]_E) = 0$, so $\chi_A(A) = 0$. $\square$

Lemma 9.6 (Primary decomposition). *Let*

$$p_A(t) = \prod_{i=1}^s (t - \lambda_i)^{m_i}$$

be the characteristic polynomial of A , with distinct λ_i . Then

$$V = G_{\lambda_1}(A) \oplus \dots \oplus G_{\lambda_s}(A).$$

Proof. Choose N large enough that $G_{\lambda_i}(A) = \text{Ker}(A - \lambda_i I)^N$ for every i . The polynomials

$$q_i(t) = (t - \lambda_i)^N$$

are pairwise coprime. Hence the polynomials

$$Q_i(t) = \prod_{j \neq i} q_j(t)$$

have greatest common divisor 1. By Bezout's identity in $\mathbb{K}[t]$, there exist polynomials $r_i(t)$ such that

$$\sum_{i=1}^s r_i(t) Q_i(t) = 1.$$

Evaluating at A gives

$$I = \sum_i r_i(A) Q_i(A).$$

For any $v \in V$, set $v_i = r_i(A) Q_i(A) v$. Then $v = \sum_i v_i$. Also,

$$(A - \lambda_i I)^N v_i = q_i(A) r_i(A) Q_i(A) v = r_i(A) \prod_{j=1}^s q_j(A) v.$$

If $N \geq \max_i m_i$, then $p_A(t)$ divides $\prod_j q_j(t)$. Hence $\prod_j q_j(t) = p_A(t) s(t)$ for some polynomial $s(t)$, and by Cayley-Hamilton, $\prod_j q_j(A) = p_A(A) s(A) = 0$. Thus $v_i \in G_{\lambda_i}(A)$, proving that the sum of the generalized eigenspaces is all of V .

To prove directness, suppose $\sum_i v_i = 0$ with $v_i \in G_{\lambda_i}(A)$. Fix k . Let

$$R_k(t) = \prod_{i \neq k} (t - \lambda_i)^N.$$

For $i \neq k$, $R_k(A)v_i = 0$, because R_k contains $(t - \lambda_i)^N$. Hence

$$0 = R_k(A) \sum_i v_i = R_k(A)v_k.$$

On $G_{\lambda_k}(A)$, each $A - \lambda_i I$ with $i \neq k$ is invertible: indeed,

$$A - \lambda_i I = (A - \lambda_k I) + (\lambda_k - \lambda_i)I,$$

where $A - \lambda_k I$ is nilpotent on $G_{\lambda_k}(A)$ and $\lambda_k - \lambda_i \neq 0$; the inverse is a finite geometric series. Thus $R_k(A)$ is invertible on $G_{\lambda_k}(A)$, so $v_k = 0$. Since k was arbitrary, all $v_i = 0$. $\square$

Lemma 9.7 (Jordan form for nilpotent operators). *Let $N \in \text{End}(V)$ be nilpotent. Then there exists a basis of V consisting of chains*

$$e, Ne, N^2e, \dots, N^{r-1}e$$

where $N^r e = 0$. In the reversed order

$$N^{r-1}e, \dots, Ne, e,$$

the matrix of N on each chain is a Jordan block $J_r(0)$.

Proof. Let r be the nilpotency index, so $N^r = 0$ and $N^{r-1} \neq 0$. Put $K_j = \text{Ker } N^j$ with $K_0 = 0$. The quotients

$$K_j / (K_{j-1} + NK_{j+1})$$

measure how many chains must start at height j . Choose a basis $\mathcal{C}_j$ in each such quotient and lift its elements to vectors in K_j . For each lifted vector $e \in \mathcal{C}_j$, form the chain

$$e, Ne, \dots, N^{j-1}e.$$

We claim that all vectors in all these chains form a basis of V .

First, they are linearly independent. Suppose a linear relation among all chain vectors exists. Apply a maximal power of N that does not kill the relation. This gives a nontrivial relation among top nonzero chain vectors of some fixed height. Passing to the quotient $K_j / (K_{j-1} + NK_{j+1})$ contradicts the chosen independence of the corresponding lifted classes. More explicitly, the terms coming from longer chains lie in NK_{j+1} , and the terms from shorter chains lie in K_{j-1} ; hence the quotient sees only the coefficients of the chosen representatives of height j , so they must vanish. Descending on j kills all coefficients.

Second, they span V . We prove by descending induction on j that K_j is contained in the span of the constructed chains together with NK_{j+1} . The quotient basis property gives

$$K_j = K_{j-1} + NK_{j+1} + \text{span}(\mathcal{C}_j).$$

By induction, K_{j-1} is already spanned by lower parts of constructed chains, and NK_{j+1} is spanned by images of higher-chain vectors. Adding $\mathcal{C}_j$ gives K_j . For $j = r$, $K_r = V$, so the chains span all of V .

Thus the chain vectors form a basis. In the order $N^{j-1}e, \dots, Ne, e$, the operator sends each vector to the previous one and sends the first vector to 0, giving the nilpotent Jordan block with zeros on the diagonal and ones on the superdiagonal. $\square$

Theorem 9.8 (Jordan canonical form). *Let $A \in \text{End}(V)$ and assume $\mathbb{K}$ is algebraically closed. Then there exists a basis of V in which the matrix of A is a block diagonal Jordan matrix:*

$$[A]_J = \text{diag}(J_{r_1}(\lambda_1), \dots, J_{r_k}(\lambda_k)).$$

The multiset of Jordan block sizes for each eigenvalue is uniquely determined by A .

Proof. By the primary decomposition lemma,

$$V = \bigoplus_{\lambda} G_{\lambda}(A).$$

Each generalized eigenspace is invariant under A . On $G_{\lambda}(A)$, define

$$N_{\lambda} = (A - \lambda I)|_{G_{\lambda}(A)}.$$

This operator is nilpotent by definition of generalized eigenspace. By the nilpotent Jordan lemma, choose a Jordan chain basis for N_{λ} . In that basis, N_{λ} has nilpotent Jordan blocks $J_r(0)$, so

$$A|_{G_{\lambda}(A)} = \lambda I + N_{\lambda}$$

has blocks $J_r(\lambda)$. Combining the bases over all eigenvalues gives a basis of V and hence a block diagonal Jordan matrix.

For uniqueness, observe that for fixed λ , the dimensions

$$d_k(\lambda) = \dim \operatorname{Ker}(A - \lambda I)^k$$

are basis-independent. In a Jordan block of size r , the contribution to $d_k(\lambda)$ is $\min(k, r)$. Hence

$$d_k - d_{k-1} = \#\{\text{Jordan blocks for } \lambda \text{ of size at least } k\}.$$

Therefore the number of blocks of size exactly k is

$$(d_k - d_{k-1}) - (d_{k+1} - d_k) = 2d_k - d_{k-1} - d_{k+1},$$

which is determined by A . Thus the block sizes are uniquely determined, up to permutation of blocks. $\square$

10. Riesz-Fischer theorem and adjoint operators

Core idea. In an inner product space, vectors and continuous linear functionals can be identified; the adjoint is the operator obtained by moving an operator from one side of the inner product to the other.

Definition 10.1 (Dual space). The *dual space* of V is

$$V^* = \operatorname{Hom}(V, \mathbb{K}),$$

the vector space of all linear functionals on V .

Theorem 10.2 (Riesz-Fischer representation, finite-dimensional form). *Let V be a finite-dimensional inner product space. For every linear functional $\varphi \in V^*$, there exists a unique vector $y \in V$ such that*

$$\varphi(x) = \langle x, y \rangle \quad \text{for all } x \in V.$$

Proof. Choose an orthonormal basis $(e_1, \dots, e_n)$ of V . For $x = \sum_i x_i e_i$, linearity of φ gives

$$\varphi(x) = \sum_i x_i \varphi(e_i).$$

Set

$$y = \sum_i \overline{\varphi(e_i)} e_i$$

with the convention that complex conjugation is trivial over $\mathbb{R}$. Then

$$\langle x, y \rangle = \left\langle \sum_i x_i e_i, \sum_j \overline{\varphi(e_j)} e_j \right\rangle = \sum_{i,j} x_i \varphi(e_j) \langle e_i, e_j \rangle = \sum_i x_i \varphi(e_i) = \varphi(x).$$

Existence is proved.

For uniqueness, suppose $\langle x, y \rangle = \langle x, z \rangle$ for all x . Then $\langle x, y - z \rangle = 0$ for all x . Taking $x = y - z$ gives $\|y - z\|^2 = 0$, hence $y = z$. $\square$

Remark 10.3 (Hilbert-space version). In an infinite-dimensional Hilbert space, the same statement holds for every continuous linear functional and is usually called the Riesz representation theorem. Finite-dimensional spaces need no separate continuity assumption because every linear functional is continuous for the norm induced by the inner product.

Definition 10.4 (Adjoint operator). Let V, W be finite-dimensional inner product spaces and let $A : V \rightarrow W$ be linear. The *adjoint* of A is the unique linear map $A^* : W \rightarrow V$ such that

$$\langle Av, w \rangle_W = \langle v, A^*w \rangle_V \quad \text{for all } v \in V, w \in W.$$

Theorem 10.5 (Existence and uniqueness of the adjoint). *Every linear map $A : V \rightarrow W$ between finite-dimensional inner product spaces has a unique adjoint $A^* : W \rightarrow V$.*

Proof. Fix $w \in W$. The map

$$\varphi_w : V \rightarrow \mathbb{K}, \quad \varphi_w(v) = \langle Av, w \rangle_W$$

is linear in v . By Riesz-Fischer, there is a unique vector $z \in V$ such that

$$\varphi_w(v) = \langle v, z \rangle_V \quad \text{for all } v.$$

Define $A^*w = z$. This gives the defining identity.

We must prove A^* is linear. Let $w_1, w_2 \in W$ and $\alpha, \beta \in \mathbb{K}$. For every $v \in V$,

$$\langle v, A^*(\alpha w_1 + \beta w_2) \rangle = \langle Av, \alpha w_1 + \beta w_2 \rangle = \bar{\alpha} \langle Av, w_1 \rangle + \bar{\beta} \langle Av, w_2 \rangle.$$

Since the inner product is conjugate-linear in the second argument,

$$\bar{\alpha} \langle Av, w_1 \rangle + \bar{\beta} \langle Av, w_2 \rangle = \langle v, \alpha A^*w_1 + \beta A^*w_2 \rangle.$$

Thus the two representing vectors are equal by uniqueness in Riesz-Fischer:

$$A^*(\alpha w_1 + \beta w_2) = \alpha A^*w_1 + \beta A^*w_2.$$

Uniqueness of the whole operator follows because each vector A^*w is uniquely determined by the representing property. $\square$

Theorem 10.6 (Algebraic properties of adjoints). *For linear maps of compatible finite-dimensional inner product spaces:*

- (i) $(A + B)^* = A^* + B^*$;
- (ii) $(\lambda A)^* = \bar{\lambda} A^*$;
- (iii) $(BA)^* = A^* B^*$;
- (iv) $(A^*)^* = A$;

$$(v) \ I^* = I.$$

Proof. (i) For all v, w ,

$$\langle (A + B)v, w \rangle = \langle Av, w \rangle + \langle Bv, w \rangle = \langle v, A^*w \rangle + \langle v, B^*w \rangle = \langle v, (A^* + B^*)w \rangle.$$

By uniqueness of the adjoint, $(A + B)^* = A^* + B^*$.

(ii) For all v, w ,

$$\langle (\lambda A)v, w \rangle = \lambda \langle Av, w \rangle = \lambda \langle v, A^*w \rangle = \langle v, \bar{\lambda}A^*w \rangle.$$

Thus $(\lambda A)^* = \bar{\lambda}A^*$.

(iii) If $A : U \rightarrow V$ and $B : V \rightarrow W$, then

$$\langle BAu, w \rangle = \langle Au, B^*w \rangle = \langle u, A^*B^*w \rangle,$$

so $(BA)^* = A^*B^*$.

(iv) Since

$$\langle Av, w \rangle = \langle v, A^*w \rangle,$$

conjugate symmetry gives equivalently

$$\langle A^*w, v \rangle = \langle w, Av \rangle.$$

Thus the adjoint of A^* is A .

(v) $\langle Iv, w \rangle = \langle v, w \rangle = \langle v, Iw \rangle$, so $I^* = I$. □

Proposition 10.7 (Matrix of the adjoint). *If E and F are orthonormal bases of V and W , and $M = [A]_{F \leftarrow E}$, then*

$$[A^*]_{E \leftarrow F} = M^*,$$

where $M^* = \overline{M}^T$ is the conjugate transpose.

Proof. Let $M = (m_{ij})$, so

$$Ae_j = \sum_i m_{ij} f_i.$$

The coefficient of e_j in A^*f_i is

$$\langle A^*f_i, e_j \rangle = \overline{\langle e_j, A^*f_i \rangle} = \overline{\langle Ae_j, f_i \rangle} = \overline{m_{ij}}.$$

Hence the matrix entry in row j , column i of $[A^*]_{E \leftarrow F}$ is $\overline{m_{ij}}$, which is exactly the conjugate transpose of M . □

11. Kernel-cokernel theorem and Fredholm alternative

Core idea. Failure of solvability of $Ax = b$ is detected exactly by vectors orthogonal to the image, equivalently by the kernel of the adjoint.

Definition 11.1 (Cokernel). For a linear map $A : V \rightarrow W$, the *cokernel* is the quotient vector space

$$\text{coker } A = W / \text{Im } A.$$

Its dimension is

$$\dim \text{coker } A = \dim W - \dim \text{Im } A.$$

Theorem 11.2 (Kernel and cokernel theorem). *Let $A : V \rightarrow W$ be linear between finite-dimensional inner product spaces. Then*

$$(\operatorname{Im} A)^\perp = \operatorname{Ker} A^*, \quad (\operatorname{Ker} A)^\perp = \operatorname{Im} A^*.$$

Consequently,

$$W = \operatorname{Im} A \oplus \operatorname{Ker} A^*,$$

and

$$\dim \operatorname{coker} A = \dim \operatorname{Ker} A^*.$$

Proof. First,

$$w \in (\operatorname{Im} A)^\perp \iff \langle Av, w \rangle = 0 \text{ for all } v \in V \iff \langle v, A^*w \rangle = 0 \text{ for all } v \in V \iff A^*w = 0.$$

Thus $(\operatorname{Im} A)^\perp = \operatorname{Ker} A^*$.

Apply the first identity to $A^* : W \rightarrow V$:

$$(\operatorname{Im} A^*)^\perp = \operatorname{Ker}(A^*)^* = \operatorname{Ker} A.$$

Taking orthogonal complements in finite dimension gives

$$\operatorname{Im} A^* = (\operatorname{Ker} A)^\perp.$$

Indeed, for every subspace U of a finite-dimensional inner product space, $(U^\perp)^\perp = U$, by the orthogonal decomposition theorem.

The orthogonal decomposition theorem applied to $\operatorname{Im} A \subseteq W$ gives

$$W = \operatorname{Im} A \oplus (\operatorname{Im} A)^\perp = \operatorname{Im} A \oplus \operatorname{Ker} A^*.$$

Therefore

$$\dim \operatorname{Ker} A^* = \dim W - \dim \operatorname{Im} A = \dim(W / \operatorname{Im} A) = \dim \operatorname{coker} A.$$

□

Theorem 11.3 (Fredholm alternative, finite-dimensional form). *Let $A : V \rightarrow W$ be linear between finite-dimensional inner product spaces. For $b \in W$, the equation*

$$Ax = b$$

has a solution if and only if

$$\langle b, y \rangle = 0 \quad \text{for every } y \in \operatorname{Ker} A^*.$$

If a solution x_0 exists, then the set of all solutions is

$$x_0 + \operatorname{Ker} A = \{x_0 + z : z \in \operatorname{Ker} A\}.$$

In particular, if $V = W$, then either A is invertible and $Ax = b$ has a unique solution for every b , or the homogeneous equation $Ax = 0$ has a nonzero solution; in the non-invertible case, solvability of $Ax = b$ is governed by the orthogonality condition $b \perp \operatorname{Ker} A^$.*

Proof. The equation $Ax = b$ is solvable if and only if $b \in \text{Im } A$. By the kernel-cokernel theorem,

$$W = \text{Im } A \oplus \text{Ker } A^*.$$

Therefore $b \in \text{Im } A$ if and only if its component in $\text{Ker } A^*$ is zero, which is equivalent to $b \perp \text{Ker } A^*$.

If x_0 is one solution, then $Ax = b$ if and only if

$$A(x - x_0) = 0,$$

which is equivalent to $x - x_0 \in \text{Ker } A$. Thus all solutions are exactly $x_0 + \text{Ker } A$.

When $V = W$, rank-nullity gives that A is injective if and only if A is surjective. Hence either A is invertible, or $\text{Ker } A \neq 0$. The last solvability statement is the already proved criterion. $\square$

12. Eigenvalues and eigenvectors

Core idea. Eigenvectors are directions preserved by an operator; on such directions the operator acts by scalar multiplication.

Definition 12.1 (Eigenvalue and eigenvector). Let $A \in \text{End}(V)$. A scalar $\lambda \in \mathbb{K}$ is an *eigenvalue* of A if there exists a nonzero vector $v \in V$ such that

$$Av = \lambda v.$$

Such a nonzero vector v is an *eigenvector* belonging to λ . The eigenspace for λ is

$$E_\lambda(A) = \text{Ker}(A - \lambda I).$$

Proposition 12.2 (Eigenspaces are subspaces). *For every $\lambda \in \mathbb{K}$, $E_\lambda(A)$ is a subspace of V .*

Proof. Since $E_\lambda(A) = \text{Ker}(A - \lambda I)$ and kernels of linear maps are subspaces, the claim follows. $\square$

Theorem 12.3 (Linear independence of eigenvectors for distinct eigenvalues). *Let $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of A , and let v_i be an eigenvector belonging to λ_i . Then $v_1, \dots, v_m$ are linearly independent.*

Proof. We prove the stronger statement that any list of nonzero vectors chosen from eigenspaces corresponding to distinct eigenvalues is independent. Suppose, for contradiction, that there is a nontrivial linear relation

$$a_1 v_1 + \dots + a_m v_m = 0$$

with the smallest possible number m of terms. Then every $a_i \neq 0$.

Apply A to the relation:

$$a_1 \lambda_1 v_1 + \dots + a_m \lambda_m v_m = 0.$$

Multiply the original relation by λ_m :

$$a_1 \lambda_m v_1 + \dots + a_m \lambda_m v_m = 0.$$

Subtracting gives

$$a_1 (\lambda_1 - \lambda_m) v_1 + \dots + a_{m-1} (\lambda_{m-1} - \lambda_m) v_{m-1} = 0.$$

This is a nontrivial relation among fewer eigenvectors, because $a_i \neq 0$ and $\lambda_i - \lambda_m \neq 0$ for $i < m$. This contradicts minimality. Hence no nontrivial relation exists. $\square$

Corollary 12.4 (Diagonalization criterion). *An operator $A \in \text{End}(V)$ is diagonalizable if and only if V has a basis consisting of eigenvectors of A .*

Proof. If V has an eigenbasis $(v_1, \dots, v_n)$ with $Av_i = \lambda_i v_i$, then the matrix of A in this basis is diagonal with diagonal entries λ_i .

Conversely, if $[A]_E$ is diagonal in a basis $E = (e_1, \dots, e_n)$, then $Ae_i = \lambda_i e_i$ for the corresponding diagonal entry λ_i . Thus every basis vector is an eigenvector. $\square$

13. Characteristic polynomial of a matrix and an operator

Core idea. The characteristic polynomial is the coordinate-independent polynomial whose roots are precisely the eigenvalues.

Definition 13.1 (Characteristic polynomial of a matrix). For $M \in \text{Mat}_n(\mathbb{K})$, the *characteristic polynomial* is

$$\chi_M(t) = \det(tI_n - M).$$

Some authors use $\det(M - tI_n)$; this differs by the factor $(-1)^n$.

Theorem 13.2 (Eigenvalues and characteristic polynomial). *A scalar $\lambda \in \mathbb{K}$ is an eigenvalue of M if and only if*

$$\chi_M(\lambda) = 0.$$

Proof. The scalar λ is an eigenvalue if and only if there exists $x \neq 0$ such that

$$Mx = \lambda x,$$

which is equivalent to

$$(\lambda I - M)x = 0.$$

This means the linear map $\lambda I - M$ has nontrivial kernel, hence is not invertible. A square matrix is not invertible if and only if its determinant is zero. Therefore

$$\det(\lambda I - M) = 0,$$

which is exactly $\chi_M(\lambda) = 0$. $\square$

Proposition 13.3 (Coefficients for 2×2 matrices). *If*

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$\chi_M(t) = t^2 - \text{tr}(M)t + \det(M).$$

Proof. Compute directly:

$$\chi_M(t) = \det \begin{pmatrix} t-a & -b \\ -c & t-d \end{pmatrix} = (t-a)(t-d) - bc = t^2 - (a+d)t + (ad-bc).$$

Since $\text{tr}(M) = a + d$ and $\det(M) = ad - bc$, the formula follows. $\square$

Proposition 13.4 (Coefficients for 3×3 matrices). *If $M = (m_{ij}) \in \text{Mat}_3(\mathbb{K})$, then*

$$\chi_M(t) = t^3 - \text{tr}(M)t^2 + s_2(M)t - \det(M),$$

where

$$s_2(M) = \det \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} + \det \begin{pmatrix} m_{11} & m_{13} \\ m_{31} & m_{33} \end{pmatrix} + \det \begin{pmatrix} m_{22} & m_{23} \\ m_{32} & m_{33} \end{pmatrix}$$

is the sum of the principal 2×2 minors.

Proof. Write

$$tI - M = \begin{pmatrix} t - m_{11} & -m_{12} & -m_{13} \\ -m_{21} & t - m_{22} & -m_{23} \\ -m_{31} & -m_{32} & t - m_{33} \end{pmatrix}.$$

Expanding the determinant, the t^3 term comes only from choosing all three diagonal t terms, so its coefficient is 1.

The t^2 terms arise by choosing two diagonal t terms and one diagonal entry $-m_{ii}$, giving coefficient

$$-(m_{11} + m_{22} + m_{33}) = -\text{tr}(M).$$

The t terms arise from choosing one diagonal t and the determinant of the complementary principal 2×2 block of $-M$. Since the complementary block is 2×2 , multiplying both rows by -1 does not change its determinant. Therefore the t coefficient is the sum of the principal 2×2 minors $s_2(M)$.

The constant term is obtained by setting $t = 0$:

$$\det(-M) = (-1)^3 \det(M) = -\det(M).$$

Combining these four parts gives the formula. □

Definition 13.5 (Characteristic polynomial of an operator). Let $A \in \text{End}(V)$ and let E be any basis of V . The characteristic polynomial of A is

$$\chi_A(t) = \chi_{[A]_E}(t) = \det(tI - [A]_E).$$

Theorem 13.6 (Basis-independence). *The characteristic polynomial of an operator is well-defined: it does not depend on the chosen basis.*

Proof. If E and F are bases of V , then

$$[A]_F = P^{-1}[A]_E P$$

for some invertible transition matrix P . Therefore

$$\chi_{[A]_F}(t) = \det(tI - P^{-1}[A]_E P) = \det(P^{-1}(tI - [A]_E)P).$$

Using multiplicativity of determinant,

$$\det(P^{-1}(tI - [A]_E)P) = \det(P^{-1}) \det(tI - [A]_E) \det(P) = \det(tI - [A]_E),$$

because $\det(P^{-1}) \det(P) = 1$. Thus the polynomial is independent of the basis. □

Corollary 13.7 (Eigenvalues of an operator). *For $A \in \text{End}(V)$ and $\lambda \in \mathbb{K}$, λ is an eigenvalue of A if and only if*

$$\chi_A(\lambda) = 0.$$

Proof. Choose a basis E . A vector $v \neq 0$ satisfies $Av = \lambda v$ if and only if its coordinate column $x = [v]_E \neq 0$ satisfies

$$[A]_E x = \lambda x.$$

By the matrix theorem, this is equivalent to

$$\det(\lambda I - [A]_E) = 0,$$

that is, $\chi_A(\lambda) = 0$. □

Proposition 13.8 (Trace and determinant are operator invariants). *The coefficients of t^{n-1} and t^0 in $\chi_A(t)$ are basis-independent. In particular,*

$$\chi_A(t) = t^n - \text{tr}(A)t^{n-1} + \cdots + (-1)^n \det(A),$$

where $\text{tr}(A)$ and $\det(A)$ are defined as the trace and determinant of any matrix representation of A .

Proof. The characteristic polynomial is basis-independent. Its coefficients are therefore basis-independent. For any matrix representative $M = [A]_E$, direct expansion of $\det(tI - M)$ gives leading term t^n , coefficient of t^{n-1} equal to $-\text{tr}(M)$, and constant term $\det(-M) = (-1)^n \det(M)$. Since these coefficients do not depend on E , neither do $\text{tr}(M)$ and $\det(M)$. □

Minimal checklist for oral exam use

1. Always state the field $\mathbb{K}$ and whether the space is finite-dimensional.
2. Distinguish a vector from its coordinate column.
3. Write transition matrices with direction: $P_{E \leftarrow F}$ sends F -coordinates to E -coordinates.
4. For operator matrices under a basis change, use similarity: $M' = P^{-1}MP$.
5. In inner product problems, specify the convention for complex linearity.
6. Orthogonal projection is characterized by $v - P_U v \in U^\perp$ and by the nearest-point property.
7. The adjoint is defined by $\langle Av, w \rangle = \langle v, A^*w \rangle$.
8. The Fredholm alternative is the solvability criterion $b \perp \text{Ker } A^*$.
9. Eigenvectors with distinct eigenvalues are linearly independent.
10. The characteristic polynomial is basis-independent because similar matrices have the same determinant $\det(tI - M)$.

How to use these notes

The notes rewrite the requested linear algebra tickets in a structural style. The central principle is that a linear operator is not merely a matrix. A linear operator is a representation of the polynomial algebra $\mathbb{F}[t]$: the variable t acts by the operator. Cyclic subspaces, annihilating polynomials, Frobenius form, Jordan form, and functions of matrices become consequences of this one idea.

Unless explicitly stated otherwise, vector spaces are finite-dimensional over a field $\mathbb{F}$. Inner-product sections are over $\mathbb{C}$ or $\mathbb{R}$. For Lax–Milgram, H denotes a Hilbert space. Every additional structural theorem used in the text is stated and proved, at least in the finite-dimensional form needed for the tickets.

1 Structural backbone: operators as $\mathbb{F}[t]$ -modules

Core idea. The modern classification of one linear operator is the classification of a finite-dimensional torsion module over the PID $\mathbb{F}[t]$.

Definition 1.1. Let $A \in \text{End}_{\mathbb{F}}(V)$. Define an $\mathbb{F}[t]$ -module structure on V by

$$p(t) \cdot v := p(A)v.$$

This module is denoted V_A .

Proposition 1.2. A subspace $W \subset V$ is A -invariant if and only if W is an $\mathbb{F}[t]$ -submodule of V_A .

Proof. If W is A -invariant, then $Aw \in W$ for all $w \in W$. Therefore $A^k w \in W$ for every $k \geq 0$, and hence $p(A)w \in W$ for every polynomial p . Thus W is an $\mathbb{F}[t]$ -submodule. Conversely, if W is an $\mathbb{F}[t]$ -submodule, then $t \cdot w = Aw \in W$, so W is A -invariant. $\square$

Definition 1.3. For $v \in V$, the cyclic subspace generated by v is

$$Z_A(v) = \mathbb{F}[t]v = \text{span}\{v, Av, A^2v, \dots\}.$$

The annihilator of v is

$$\text{Ann}_A(v) = \{p \in \mathbb{F}[t] : p(A)v = 0\}.$$

The annihilator of A , also called the annihilating ideal of A , is

$$\text{Ann}(A) = \{p \in \mathbb{F}[t] : p(A) = 0\}.$$

Lemma 1.4 (The polynomial ring is a principal ideal domain). *Every ideal $I \subset \mathbb{F}[t]$ is either (0) or (q) for some nonzero polynomial $q \in \mathbb{F}[t]$. If $I \neq 0$, then q can be chosen monic and is unique.*

Proof. If $I = 0$, there is nothing to prove. If $I \neq 0$, choose $q \in I$ of minimal degree and multiply by a scalar so that q is monic. For any $p \in I$, Euclidean division gives $p = sq + r$, where either $r = 0$ or $\deg r < \deg q$. Since $p, sq \in I$, we have $r \in I$. Minimality of q forces $r = 0$. Hence $p \in (q)$. Therefore $I = (q)$. If $(q) = (q')$ and both are monic, then $q \mid q'$ and $q' \mid q$, so $q = cq'$ for a unit $c \in \mathbb{F}^\times$; monicity gives $c = 1$. $\square$

Definition 1.5. The monic generator of $\text{Ann}(A)$ is the minimal polynomial of A , denoted $m_A(t)$. The monic generator of $\text{Ann}_A(v)$ is the local minimal polynomial of v , denoted $m_v(t)$.

Lemma 1.6 (Cyclic module lemma). *Let $v \in V$. The map*

$$\Phi_v : \mathbb{F}[t] \rightarrow Z_A(v), \quad p \mapsto p(A)v$$

is a surjective $\mathbb{F}[t]$ -module homomorphism with kernel $\text{Ann}_A(v) = (m_v)$. Hence

$$Z_A(v) \cong \mathbb{F}[t]/(m_v).$$

If $\deg m_v = r$, then

$$v, Av, \dots, A^{r-1}v$$

is a basis of $Z_A(v)$.

Proof. Surjectivity follows from the definition of $Z_A(v)$. The kernel consists exactly of all polynomials p with $p(A)v = 0$, hence it is $\text{Ann}_A(v) = (m_v)$. The first isomorphism theorem for modules gives $Z_A(v) \cong \mathbb{F}[t]/(m_v)$. If $\deg m_v = r$, every residue class modulo m_v has a unique representative of degree $< r$. Therefore $1, t, \dots, t^{r-1}$ form an $\mathbb{F}$ -basis of $\mathbb{F}[t]/(m_v)$, and their images are $v, Av, \dots, A^{r-1}v$. $\square$

Definition 1.7. For a monic polynomial

$$p(t) = t^r + a_{r-1}t^{r-1} + \dots + a_1t + a_0,$$

the Frobenius companion matrix used in these notes is

$$C(p) = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{r-1} \end{pmatrix}.$$

This convention is the transpose of another common convention. The two conventions are similar after reversing the cyclic basis.

Proposition 1.8 (Cyclic block). *If $Z_A(v)$ has local minimal polynomial p of degree r , then in the cyclic basis*

$$A^{r-1}v, A^{r-2}v, \dots, Av, v$$

the matrix of $A|_{Z_A(v)}$ is $C(p)$.

Proof. The relation $p(A)v = 0$ gives

$$A^r v = -a_{r-1}A^{r-1}v - \dots - a_1Av - a_0v.$$

With the reversed cyclic basis $e_1 = A^{r-1}v, \dots, e_r = v$, the operator sends $e_2 \mapsto e_1$, $e_3 \mapsto e_2$, and so on, while $e_1 = A^{r-1}v$ is sent to the displayed linear combination. This is the companion matrix above. $\square$

Theorem 1.9 (Smith normal form over the polynomial ring). *Let M be an $m \times n$ matrix with entries in $\mathbb{F}[t]$. There exist invertible matrices $P \in GL_m(\mathbb{F}[t])$ and $Q \in GL_n(\mathbb{F}[t])$ such that*

$$PMQ = \text{diag}(d_1, \dots, d_r, 0, \dots, 0),$$

where the nonzero d_i are monic and

$$d_1 \mid d_2 \mid \dots \mid d_r.$$

Proof. The proof is the Euclidean algorithm applied to matrices. Choose a nonzero entry of minimal degree and move it to the upper-left corner using row and column swaps. Use Euclidean division to reduce every entry in the first row and first column modulo that entry. If some remainder is nonzero, it has smaller degree; move it to the upper-left corner and repeat. The degree strictly decreases, so the process stops. At that point the upper-left entry divides all entries in the first row and first column. If it fails to divide some other entry, adding the row and column containing that entry to the first row and column produces a smaller remainder, contradiction to the stopping condition. Hence the upper-left entry divides every entry of the matrix. Now clear the rest of the first row and column by elementary row and column operations. This gives a block

$$\begin{pmatrix} d_1 & 0 \\ 0 & M_1 \end{pmatrix}$$

with d_1 dividing every entry of M_1 . Apply induction to M_1 . If two adjacent diagonal entries fail the divisibility condition, replace the corresponding 2×2 diagonal block by an equivalent one using Bezout identities in the PID $\mathbb{F}[t]$; this replaces the pair by their gcd and lcm. Repeating yields $d_i \mid d_{i+1}$. All operations used are multiplication by invertible elementary matrices over $\mathbb{F}[t]$. $\square$

Theorem 1.10 (Structure theorem for finite-dimensional polynomial modules). *Let V_A be a finite-dimensional $\mathbb{F}[t]$ -module coming from $A \in \text{End}_{\mathbb{F}}(V)$. Then*

$$V_A \cong \bigoplus_{i=1}^s \mathbb{F}[t]/(q_i(t)),$$

where the q_i are monic nonconstant polynomials satisfying

$$q_1 \mid q_2 \mid \cdots \mid q_s.$$

The polynomials q_i are uniquely determined. They are called the invariant factors of A .

Proof. Choose an $\mathbb{F}$ -basis of V of size n . The module V_A is generated by these n vectors over $\mathbb{F}[t]$. The relations are encoded by the $\mathbb{F}[t]$ -linear map

$$\mathbb{F}[t]^n \xrightarrow{tI-A} \mathbb{F}[t]^n.$$

Indeed, quotienting by the relations $(tI - A)e_i = 0$ imposes $t \cdot v_i = Av_i$, so

$$V_A \cong \text{coker}(tI - A).$$

Apply Smith normal form to $tI - A$. Since V is finite-dimensional over $\mathbb{F}$, every vector is killed by some nonzero polynomial: the vectors $v, Av, \dots, A^n v$ are linearly dependent. Hence the module is torsion, so no free summand remains. Thus

$$V_A \cong \bigoplus_i \mathbb{F}[t]/(q_i).$$

The divisibility chain is part of Smith normal form. Uniqueness follows from uniqueness of Smith normal form: the determinantal ideals generated by all $k \times k$ minors are invariant under invertible row and column operations, and these ideals determine the products $q_1 \cdots q_k$, hence each q_i . $\square$

Remark 1.11. The structural theorem is the source of Frobenius normal form. Jordan form is obtained from it after factoring the invariant factors into powers of linear polynomials over an algebraically closed field.

Practice problems.

1. Prove directly that $Z_A(v)$ is the smallest A -invariant subspace containing v .
2. For $p(t) = t^3 + 2t^2 - t + 5$, write $C(p)$ and compute $p(C(p))$.
3. Let A be nilpotent with $A^3 = 0$. What can the cyclic modules $\mathbb{F}[t]v$ look like?

2 Linear independence of eigenvectors

Core idea. Eigenvectors with distinct eigenvalues are independent because the operator separates coefficients by different scalar actions.

Definition 2.1. Let $A \in \text{End}(V)$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue if $\text{Ker}(A - \lambda I) \neq 0$. A nonzero vector in $\text{Ker}(A - \lambda I)$ is an eigenvector. The eigenspace is

$$E_\lambda(A) = \text{Ker}(A - \lambda I).$$

Theorem 2.2 (Linear independence of eigenvectors). *Let $v_1, \dots, v_m$ be eigenvectors of A corresponding to pairwise distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Then $v_1, \dots, v_m$ are linearly independent.*

Proof. Proceed by induction on m . The case $m = 1$ is clear. Suppose

$$\alpha_1 v_1 + \dots + \alpha_m v_m = 0.$$

Apply A and subtract λ_m times the original relation:

$$\alpha_1(\lambda_1 - \lambda_m)v_1 + \dots + \alpha_{m-1}(\lambda_{m-1} - \lambda_m)v_{m-1} = 0.$$

By induction, $v_1, \dots, v_{m-1}$ are independent. Since $\lambda_i \neq \lambda_m$, we get $\alpha_i = 0$ for $i < m$. Then $\alpha_m v_m = 0$, so $\alpha_m = 0$. $\square$

Corollary 2.3. *The sum of eigenspaces for distinct eigenvalues is direct:*

$$E_{\lambda_1} \oplus \dots \oplus E_{\lambda_m}.$$

If $\dim V = n$ and A has n distinct eigenvalues, then A is diagonalizable.

Proof. Choose a basis in each eigenspace. Any finite linear dependence among all chosen vectors splits into a dependence among eigenvectors from distinct eigenvalues after grouping by eigenspaces; applying the theorem to one nonzero vector from each nonzero group gives all groups zero. The diagonalizability claim follows because n independent eigenvectors form a basis. $\square$

Oral-exam minimum.

- State eigenvalue, eigenvector, eigenspace.
- Prove the independence theorem without determinants.
- Explain why distinct eigenvalues are sufficient but not necessary for diagonalizability.

Practice problems.

1. Prove that if A has m distinct eigenvalues, then $\dim V \geq m$.
2. Give a diagonalizable matrix with one eigenvalue only.
3. Let $p \in \mathbb{F}[t]$. Show that if $Av = \lambda v$, then $p(A)v = p(\lambda)v$.

3 Invariant and cyclic subspaces

Core idea. Invariant subspaces are submodules; cyclic subspaces are modules generated by one vector.

Definition 3.1. A subspace $W \subset V$ is operator-invariant, or A -invariant, if $A(W) \subset W$. For $v \in V$, the cyclic subspace generated by v is

$$Z_A(v) = \mathbb{F}[t]v = \text{span}\{v, Av, A^2v, \dots\}.$$

Proposition 3.2. $Z_A(v)$ is the smallest A -invariant subspace containing v .

Proof. It is invariant because

$$A(A^k v) = A^{k+1} v \in Z_A(v).$$

It contains v . If W is invariant and contains v , then by induction it contains $A^k v$ for every $k \geq 0$, hence contains their span. $\square$

Proposition 3.3 (Restriction and quotient). *If $W \subset V$ is A -invariant, then A induces linear operators*

$$A|_W : W \rightarrow W, \quad \bar{A} : V/W \rightarrow V/W, \quad \bar{A}(v + W) = Av + W.$$

Proof. The restriction is clear. For the quotient, if $v + W = v' + W$, then $v - v' \in W$. Since W is invariant, $A(v - v') \in W$, so $Av + W = Av' + W$. Hence $\bar{A}$ is well-defined and linear. $\square$

Proposition 3.4 (Block form). *If W is A -invariant and a basis of W is extended to a basis of V , then the matrix of A has block form*

$$\begin{pmatrix} B & C \\ 0 & D \end{pmatrix}$$

up to the convention that coordinates of W are written first. The block B represents $A|_W$, and D represents the induced operator on V/W .

Proof. The first basis vectors span W . Images of vectors in W remain in W , so the lower-left block is zero. The action on the quotient ignores the W -coordinates, giving the block D . $\square$

Oral-exam minimum.

- Translate invariant subspace into $\mathbb{F}[t]$ -module language.
- Define cyclic subspace and prove its minimality.
- Explain restriction and quotient operators.

Practice problems.

1. Find all invariant subspaces of a single Jordan block $J_3(0)$.
2. Prove that $\text{Ker } p(A)$ and $\text{Im } p(A)$ are A -invariant.
3. If $V = Z_A(v)$, prove that A is represented by a companion matrix in a suitable basis.

4 Annihilating polynomials and the minimal polynomial

Core idea. Polynomial identities of an operator form an ideal. The minimal polynomial is the monic generator of that ideal.

Definition 4.1. A polynomial $p \in \mathbb{F}[t]$ is an annihilating polynomial for A if

$$p(A) = 0.$$

The set of all such polynomials is

$$\text{Ann}(A) = \{p \in \mathbb{F}[t] : p(A) = 0\}.$$

The minimal polynomial m_A is the monic generator of $\text{Ann}(A)$.

Proposition 4.2. $\text{Ann}(A)$ is a nonzero ideal of $\mathbb{F}[t]$. Hence m_A exists.

Proof. If $p(A) = q(A) = 0$, then $(p + q)(A) = 0$. If $r \in \mathbb{F}[t]$, then $(rp)(A) = r(A)p(A) = 0$. Hence $\text{Ann}(A)$ is an ideal. It is nonzero because $I, A, A^2, \dots, A^{n^2}$ are linearly dependent in the n^2 -dimensional vector space $\text{End}(V)$, so some nonzero polynomial annihilates A . Since $\mathbb{F}[t]$ is a PID, the ideal has a monic generator. $\square$

Proposition 4.3 (Divisibility property). *For $p \in \mathbb{F}[t]$,*

$$p(A) = 0 \iff m_A \mid p.$$

Proof. Since $\text{Ann}(A) = (m_A)$, a polynomial lies in $\text{Ann}(A)$ exactly when it is a multiple of m_A . $\square$

Proposition 4.4 (Spectrum and minimal polynomial). *If $\lambda \in \mathbb{F}$ is an eigenvalue of A , then $m_A(\lambda) = 0$. Conversely, if $\mathbb{F}$ is algebraically closed and $m_A(\lambda) = 0$, then λ is an eigenvalue.*

Proof. If $Av = \lambda v$ with $v \neq 0$, then $0 = m_A(A)v = m_A(\lambda)v$, so $m_A(\lambda) = 0$. Conversely, over an algebraically closed field, if $m_A(\lambda) = 0$, write $m_A = (t - \lambda)q$. If $A - \lambda I$ were invertible, then $q(A) = 0$, contradicting the minimality of m_A . Therefore $A - \lambda I$ is not invertible and λ is an eigenvalue. $\square$

Proposition 4.5 (Minimal polynomial of a direct sum). *If $V = V_1 \oplus \dots \oplus V_s$ is a direct sum of invariant subspaces and $A_i = A|_{V_i}$, then*

$$m_A = \text{lcm}(m_{A_1}, \dots, m_{A_s}).$$

Proof. A polynomial p annihilates A if and only if it annihilates every restriction A_i . This is equivalent to $m_{A_i} \mid p$ for every i , hence to $\text{lcm}(m_{A_i}) \mid p$. The monic generator is the lcm. $\square$

Oral-exam minimum.

- Define annihilating polynomial and minimal polynomial.
- Prove the ideal property and divisibility property.
- Explain how eigenvalues are detected by m_A .

Practice problems.

1. Compute the minimal polynomial of $\text{diag}(1, 1, 2, 3)$.
2. Compute the minimal polynomial of a Jordan matrix $J_3(0) \oplus J_2(0)$.
3. Prove that A is invertible if and only if $m_A(0) \neq 0$.

5 Frobenius normal form

Core idea. Frobenius normal form is the matrix form of the invariant-factor decomposition of the $\mathbb{F}[t]$ -module V_A .

Definition 5.1. A Frobenius block is a companion matrix $C(q)$ of a monic polynomial q . A matrix is in Frobenius normal form if it is block diagonal

$$C(q_1) \oplus \dots \oplus C(q_s)$$

where

$$q_1 \mid q_2 \mid \dots \mid q_s.$$

The polynomials q_i are the invariant factors.

Theorem 5.2 (Frobenius normal form). *For every $A \in \text{End}_{\mathbb{F}}(V)$ there exists a basis in which*

$$[A] = C(q_1) \oplus \dots \oplus C(q_s),$$

where $q_1, \dots, q_s$ are the invariant factors of V_A . This form is unique up to the invariant factors.

Proof. By the structure theorem for finite-dimensional $\mathbb{F}[t]$ -modules,

$$V_A \cong \bigoplus_{i=1}^s \mathbb{F}[t]/(q_i), \quad q_1 \mid \cdots \mid q_s.$$

Each summand is cyclic. Choose in the i -th summand the basis corresponding to

$$t^{r_i-1}, t^{r_i-2}, \dots, t, 1, \quad r_i = \deg q_i.$$

Multiplication by t in this basis is exactly the companion matrix $C(q_i)$. Taking the union of these cyclic bases gives a basis of V in which A is block diagonal with Frobenius blocks. Uniqueness follows from uniqueness of invariant factors in the structure theorem. $\square$

Corollary 5.3. *If the invariant factors of A are $q_1 \mid \cdots \mid q_s$, then*

$$m_A = q_s, \quad \chi_A = q_1 q_2 \cdots q_s.$$

Proof. On the block $C(q_i)$, the minimal polynomial is q_i , and the characteristic polynomial is q_i . For a direct sum, the minimal polynomial is the lcm of the block minimal polynomials, which is q_s , and the characteristic polynomial is the product of the block characteristic polynomials. $\square$

Oral-exam minimum.

- State Frobenius normal form through invariant factors.
- Explain why it exists over arbitrary fields.
- Relate invariant factors to m_A and χ_A .

Practice problems.

1. List all possible Frobenius forms of a 4-dimensional operator with minimal polynomial $(t-1)^2$.
2. Prove that an operator is cyclic if and only if its Frobenius form has one block.
3. Show that if $m_A = \chi_A$, then A is cyclic.

6 Cayley–Hamilton theorem

Core idea. The characteristic polynomial annihilates the operator because it annihilates each cyclic Frobenius block.

Lemma 6.1. *For a companion matrix $C(p)$, one has*

$$p(C(p)) = 0, \quad \chi_{C(p)}(t) = p(t).$$

Proof. The first equality follows from the cyclic construction: $C(p)$ represents multiplication by t on $\mathbb{F}[t]/(p)$, so multiplication by $p(t)$ is zero. For the characteristic polynomial, compute $\det(tI - C(p))$ by expanding along the first column or use the recurrence obtained from the companion shape; the result is exactly

$$t^r + a_{r-1}t^{r-1} + \cdots + a_0 = p(t).$$

$\square$

Theorem 6.2 (Cayley–Hamilton). *For every $A \in \text{End}(V)$,*

$$\chi_A(A) = 0.$$

Proof. Put A into Frobenius normal form:

$$A \sim C(q_1) \oplus \cdots \oplus C(q_s).$$

The characteristic polynomial is

$$\chi_A = q_1 q_2 \cdots q_s.$$

On the block $C(q_i)$, the polynomial q_i annihilates the block. Hence the product $q_1 \cdots q_s$ annihilates every block. Therefore $\chi_A(A) = 0$ on the direct sum. Similar matrices satisfy the same polynomial identities, so $\chi_A(A) = 0$ for the original operator. $\square$

Remark 6.3. This proof shows more than Cayley–Hamilton: the minimal polynomial divides the characteristic polynomial.

Oral-exam minimum.

- State Cayley–Hamilton exactly.
- Prove it using Frobenius form.
- Explain why $m_A \mid \chi_A$.

Practice problems.

1. Use Cayley–Hamilton to compute A^{-1} for a matrix with $\chi_A(t) = t^3 - 2t + 1$ and $\det A \neq 0$.
2. Prove that every 2×2 matrix satisfies $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$.
3. Give an example where $m_A \neq \chi_A$.

7 Jordan normal form and Jordan bases

Core idea. Over an algebraically closed field, invariant factors split into powers of linear factors. Jordan form is the cyclic decomposition of nilpotent parts on generalized eigenspaces.

Definition 7.1. Assume $\mathbb{F}$ is algebraically closed. The generalized eigenspace of A corresponding to λ is

$$V_\lambda = \operatorname{Ker}(A - \lambda I)^N$$

for any sufficiently large N , for example $N = \dim V$.

Theorem 7.2 (Primary decomposition). *Let*

$$m_A(t) = \prod_{j=1}^r (t - \lambda_j)^{s_j}$$

with distinct λ_j . Then

$$V = \bigoplus_{j=1}^r V_{\lambda_j}, \quad V_{\lambda_j} = \operatorname{Ker}(A - \lambda_j I)^{s_j}.$$

Proof. The polynomials $p_j = (t - \lambda_j)^{s_j}$ are pairwise coprime. For each j , set $g_j = m_A/p_j$. The polynomials g_j have gcd 1, so Bezout gives polynomials u_j such that

$$\sum_j u_j(t) g_j(t) = 1.$$

Applying this to A gives

$$I = \sum_j u_j(A) g_j(A).$$

For any $v \in V$, define $v_j = u_j(A) g_j(A) v$. Since $p_j(A) g_j(A) = m_A(A) = 0$, we get $v_j \in \operatorname{Ker} p_j(A) = V_{\lambda_j}$, and $v = \sum_j v_j$. Thus the sum is all of V . If $w \in V_{\lambda_i} \cap \sum_{j \neq i} V_{\lambda_j}$, then both $p_i(A)w = 0$ and $\prod_{j \neq i} p_j(A)w = 0$. These two polynomials are coprime, so Bezout implies $w = 0$. Hence the sum is direct. $\square$

Definition 7.3. A Jordan chain of length r for eigenvalue λ is a sequence

$$v_1, \dots, v_r$$

such that

$$(A - \lambda I)v_1 = 0, \quad (A - \lambda I)v_k = v_{k-1} \quad (k \geq 2).$$

In the basis $v_1, \dots, v_r$, the operator is represented by a Jordan block

$$J_r(\lambda) = \lambda I + N_r,$$

where N_r has ones on the superdiagonal or subdiagonal depending on basis convention.

Theorem 7.4 (Nilpotent Jordan basis). *Let $N \in \text{End}(V)$ be nilpotent. Then V has a basis consisting of chains*

$$e, Ne, \dots, N^{r-1}e.$$

Equivalently, N has a Jordan form with zero diagonal.

Proof. Let $N^s = 0$. Define

$$K_i = \text{Ker } N^i, \quad K_0 = 0.$$

Choose complements L_i so that

$$K_i = K_{i-1} \oplus L_i$$

for $1 \leq i \leq s$, in a way compatible with the maps induced by $N : K_i/K_{i-1} \rightarrow K_{i-1}/K_{i-2}$. Start from a basis of the top quotient K_s/K_{s-1} ; lift its vectors to K_s , and attach their images under $N, N^2, \dots$. Then continue with a basis of the remaining part of K_{s-1}/K_{s-2} , and so on. At each stage, the images of new top vectors fill a complement to the chains already constructed. The resulting chain vectors span each layer K_i/K_{i-1} with the correct dimension. Hence they form a basis of V . In that basis, N shifts along chains and kills the first vector of each chain. $\square$

Theorem 7.5 (Jordan normal form). *If $\mathbb{F}$ is algebraically closed, every $A \in \text{End}_{\mathbb{F}}(V)$ has a basis in which*

$$[A] = \bigoplus_{\lambda \in \text{Spec}(A)} \bigoplus_j J_{r_{\lambda,j}}(\lambda).$$

Proof. By primary decomposition,

$$V = \bigoplus_{\lambda} V_{\lambda}.$$

On V_{λ} , write

$$A = \lambda I + N_{\lambda}, \quad N_{\lambda} = A - \lambda I.$$

The operator N_{λ} is nilpotent on V_{λ} . Apply the nilpotent Jordan basis theorem to N_{λ} . In each chain, A acts as λI plus the nilpotent shift. Combining the bases over all λ gives the Jordan basis. $\square$

Oral-exam minimum.

- Define generalized eigenspaces.
- Prove primary decomposition using Bezout.
- Explain how Jordan chains give Jordan blocks.

Practice problems.

1. Construct a Jordan basis for a nilpotent operator with $\dim \text{Ker } N = 2$, $\dim \text{Ker } N^2 = 4$, $\dim V = 5$.
2. Prove that the number of Jordan blocks for λ is $\dim \text{Ker}(A - \lambda I)$.
3. Prove that the largest Jordan block for λ has size equal to the exponent of $t - \lambda$ in m_A .

8 Polynomials and functions of Jordan cells and operators

Core idea. Functions of operators are defined by reducing functions to polynomials modulo the nilpotent data encoded by Jordan blocks.

Proposition 8.1 (Polynomial of a Jordan block). *Let $J = J_r(\lambda) = \lambda I + N$, where $N^r = 0$. For $p \in \mathbb{F}[t]$,*

$$p(J) = \sum_{k=0}^{r-1} \frac{p^{(k)}(\lambda)}{k!} N^k$$

whenever the derivatives make sense in the field, for example in characteristic zero.

Proof. Write Taylor's formula for the polynomial around λ :

$$p(t) = \sum_{k=0}^{r-1} \frac{p^{(k)}(\lambda)}{k!} (t - \lambda)^k + (t - \lambda)^r q(t).$$

Substitute $t = J$. Since $J - \lambda I = N$ and $N^r = 0$, the remainder vanishes. $\square$

Definition 8.2. Let A have minimal polynomial

$$m_A(t) = \prod_{\lambda} (t - \lambda)^{s_{\lambda}}$$

over an algebraically closed field of characteristic zero. A function f is defined on the spectral data of A if the numbers

$$f^{(k)}(\lambda), \quad 0 \leq k < s_{\lambda},$$

are specified for each eigenvalue λ . Define $f(A) = p(A)$, where p is any polynomial satisfying the Hermite interpolation conditions

$$p^{(k)}(\lambda) = f^{(k)}(\lambda), \quad 0 \leq k < s_{\lambda}.$$

Theorem 8.3 (Well-defined functional calculus). *The operator $f(A)$ above is well-defined. Moreover,*

$$\mathbb{F}[A] \cong \mathbb{F}[t]/(m_A),$$

so $p(A) = q(A)$ if and only if $p \equiv q \pmod{m_A}$.

Proof. The evaluation map

$$\text{ev}_A : \mathbb{F}[t] \rightarrow \text{End}(V), \quad p \mapsto p(A)$$

is an algebra homomorphism. Its kernel is $\text{Ann}(A) = (m_A)$. Therefore the first isomorphism theorem for algebras gives

$$\mathbb{F}[A] = \text{Im ev}_A \cong \mathbb{F}[t]/(m_A).$$

Hermite interpolation gives a polynomial p with the prescribed derivative data. If p and q have the same data, then $p - q$ has zero derivatives up to order $s_{\lambda} - 1$ at every λ , hence $m_A \mid p - q$. Thus $p(A) = q(A)$. $\square$

Corollary 8.4. *For a Jordan block $J_r(\lambda) = \lambda I + N$,*

$$f(J_r(\lambda)) = \sum_{k=0}^{r-1} \frac{f^{(k)}(\lambda)}{k!} N^k.$$

Oral-exam minimum.

- Compute $p(J_r(\lambda))$ using Taylor expansion.
- Explain why $\mathbb{F}[A] \cong \mathbb{F}[t]/(m_A)$.

- Define $f(A)$ using Hermite interpolation.

Practice problems.

1. Compute $e^{J_3(\lambda)}$.
2. Compute $(I + N)^{-1}$ for $N^4 = 0$.
3. Prove that if $p(A) = 0$, then p vanishes on the spectrum with multiplicity at least the size of each corresponding largest Jordan block.

9 Danilevsky method and the characteristic polynomial

Core idea. Danilevsky's method constructs a matrix similar to a Frobenius companion matrix; the last row then reads off the characteristic polynomial.

Definition 9.1. A matrix is in Danilevsky–Frobenius form if it has the companion shape

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -c_0 & -c_1 & -c_2 & \cdots & -c_{n-1} \end{pmatrix}.$$

Its characteristic polynomial is

$$t^n + c_{n-1}t^{n-1} + \cdots + c_1t + c_0.$$

Theorem 9.2 (Correctness of Danilevsky's method). *Suppose elementary similarity transformations reduce a matrix A to a companion matrix $C(p)$. Then*

$$\chi_A(t) = p(t).$$

Proof. Every Danilevsky step has the form

$$A \mapsto S^{-1}AS$$

for an invertible matrix S . Similar matrices have the same characteristic polynomial because

$$\det(tI - S^{-1}AS) = \det(S^{-1}(tI - A)S) = \det(tI - A).$$

After reduction to $C(p)$, the characteristic polynomial is p by the companion matrix lemma. Therefore $\chi_A = p$. $\square$

Remark 9.3 (What the algorithm does). At step k , one uses a nonzero pivot below the subdiagonal to create the row/column relations that force the entries above the final row to become those of the companion matrix. If a pivot vanishes, the algorithm splits the problem into invariant blocks, so the characteristic polynomial becomes a product of smaller characteristic polynomials.

Oral-exam minimum.

- State what a companion matrix is.
- Explain why similarity preserves the characteristic polynomial.
- Explain how Danilevsky reads off χ_A from the final matrix.

Practice problems.

1. Apply the Danilevsky idea to a 2×2 matrix and recover $t^2 - (\text{tr } A)t + \det A$.
2. Explain why a zero pivot leads to a block decomposition.
3. Show that if Danilevsky gives one companion block, then the matrix is cyclic.

10 Eigenvectors and generalized eigenvectors of a Frobenius matrix

Core idea. The companion matrix packages polynomial roots into eigenvectors. Multiple roots produce generalized eigenvector chains through derivatives.

Theorem 10.1 (Eigenvectors of a companion matrix). *Let*

$$p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$$

and let $C(p)$ be the companion matrix with ones on the superdiagonal and last row $(-a_0, \dots, -a_{n-1})$. If $p(\lambda) = 0$, then

$$u(\lambda) = (1, \lambda, \dots, \lambda^{n-1})^T$$

is an eigenvector of $C(p)$ with eigenvalue λ .

Proof. Multiplying gives

$$C(p)u(\lambda) = (\lambda, \lambda^2, \dots, \lambda^{n-1}, -a_0 - a_1\lambda - \cdots - a_{n-1}\lambda^{n-1})^T.$$

Since $p(\lambda) = 0$, the last coordinate equals λ^n . Therefore

$$C(p)u(\lambda) = \lambda u(\lambda).$$

□

Theorem 10.2 (Generalized eigenvectors by differentiation). *Let λ be a root of p of multiplicity s . Define*

$$u_j = \frac{1}{j!} \frac{d^j}{dt^j} (1, t, \dots, t^{n-1})^T \Big|_{t=\lambda}, \quad 0 \leq j < s.$$

Then

$$(C(p) - \lambda I)u_0 = 0, \quad (C(p) - \lambda I)u_j = u_{j-1} \quad (1 \leq j < s).$$

Thus $u_0, \dots, u_{s-1}$ form a Jordan chain.

Proof. For $u(t) = (1, t, \dots, t^{n-1})^T$, direct multiplication gives

$$(C(p) - tI)u(t) = -p(t)e_n.$$

Differentiate j times:

$$(C(p) - tI)u^{(j)}(t) - ju^{(j-1)}(t) = -p^{(j)}(t)e_n.$$

At $t = \lambda$, the derivatives $p^{(j)}(\lambda)$ vanish for $j < s$. Dividing by $j!$ gives

$$(C(p) - \lambda I)u_j = u_{j-1}$$

for $j \geq 1$, and the case $j = 0$ gives the eigenvector equation. □

Oral-exam minimum.

- Write eigenvectors of a companion matrix from roots of its polynomial.
- Explain why multiple roots create generalized eigenvectors.
- Know the derivative identity $(C - tI)u(t) = -p(t)e_n$.

Practice problems.

1. Find a Jordan chain for the companion matrix of $(t - 2)^3$.
2. For $p(t) = (t - 1)^2(t + 1)$, write eigenvectors and generalized eigenvectors of $C(p)$.
3. Prove that if p has no repeated roots, then $C(p)$ is diagonalizable over a splitting field of p .

11 Complexification of vector spaces and operators

Core idea. Complexification lets one use algebraic closure while remembering the original real structure.

Definition 11.1. For a real vector space V , its complexification is

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

Concretely, every element can be written as

$$v + iw, \quad v, w \in V,$$

with scalar multiplication by $a + ib$ defined by

$$(a + ib)(v + iw) = (av - bw) + i(bv + aw).$$

Proposition 11.2. If $\dim_{\mathbb{R}} V = n$, then $\dim_{\mathbb{C}} V_{\mathbb{C}} = n$. If $e_1, \dots, e_n$ is a real basis of V , it is a complex basis of $V_{\mathbb{C}}$.

Proof. Every $v + iw$ can be written uniquely as

$$\sum_j (\alpha_j + i\beta_j)e_j$$

with $v = \sum \alpha_j e_j$, $w = \sum \beta_j e_j$. Hence the same list is a complex basis. $\square$

Definition 11.3. If $A : V \rightarrow V$ is real-linear, its complexification is

$$A_{\mathbb{C}} : V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}, \quad A_{\mathbb{C}}(v + iw) = Av + iAw.$$

Proposition 11.4. The characteristic and minimal polynomials of A and $A_{\mathbb{C}}$ are the same polynomials, considered over $\mathbb{R}$ and $\mathbb{C}$ respectively.

Proof. In a real basis of V , the matrix of A is a real matrix M . In the corresponding complex basis of $V_{\mathbb{C}}$, the matrix of $A_{\mathbb{C}}$ is the same matrix M , now viewed over $\mathbb{C}$. Therefore $\det(tI - M)$ is unchanged. Also $p(A) = 0$ over $\mathbb{R}$ if and only if $p(A_{\mathbb{C}}) = 0$ for real polynomials p , because the matrices are the same. Hence the minimal polynomial is the same. $\square$

Definition 11.5. There is a conjugation map on $V_{\mathbb{C}}$:

$$\overline{v + iw} = v - iw.$$

A complex subspace $W \subset V_{\mathbb{C}}$ is defined over $\mathbb{R}$ if $\overline{W} = W$.

Oral-exam minimum.

- Define $V_{\mathbb{C}}$ and $A_{\mathbb{C}}$.
- Prove dimensions and basis compatibility.
- Explain why nonreal eigenvalues appear in conjugate pairs for real operators.

Practice problems.

1. Complexify a real rotation matrix and find its eigenvectors.
2. Prove that if A is real, then λ is an eigenvalue of $A_{\mathbb{C}}$ iff $\bar{\lambda}$ is.
3. Show that a real invariant subspace complexifies to a conjugation-invariant complex invariant subspace.

12 Real Jordan form

Core idea. A nonreal complex Jordan block and its conjugate combine into a real block built from 2×2 rotation-scaling matrices.

Definition 12.1. For $\lambda = a + ib$, $b \neq 0$, define the real 2×2 block

$$R_\lambda = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

A real Jordan block associated with λ of size r is the $2r \times 2r$ block matrix

$$J_r^{\mathbb{R}}(\lambda) = \begin{pmatrix} R_\lambda & I_2 & 0 & \cdots & 0 \\ 0 & R_\lambda & I_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & I_2 \\ 0 & 0 & 0 & \cdots & R_\lambda \end{pmatrix}.$$

Theorem 12.2 (Real Jordan form). *Every real linear operator has a real basis in which its matrix is a block diagonal sum of:*

- ordinary real Jordan blocks $J_r(\lambda)$ for real eigenvalues λ ,
- real Jordan blocks $J_r^{\mathbb{R}}(a + ib)$ for pairs of nonreal conjugate eigenvalues.

Proof. Complexify A to $A_{\mathbb{C}}$. Over $\mathbb{C}$, choose a Jordan basis. If $\lambda \in \mathbb{R}$, conjugation preserves the generalized eigenspace and one may choose chains fixed by conjugation, giving real Jordan blocks. If $\lambda = a + ib$ with $b \neq 0$, conjugation maps a Jordan chain

$$z_1, \dots, z_r$$

for λ to a chain

$$\bar{z}_1, \dots, \bar{z}_r$$

for $\bar{\lambda}$. Write

$$z_k = x_k + iy_k, \quad x_k, y_k \in V.$$

The real vectors

$$x_1, y_1, \dots, x_r, y_r$$

span the real subspace corresponding to the pair of conjugate chains. Since

$$Az_k = \lambda z_k + z_{k-1}$$

with $z_0 = 0$, taking real and imaginary parts gives

$$Ax_k = ax_k - by_k + x_{k-1}, \quad Ay_k = bx_k + ay_k + y_{k-1}.$$

This is exactly the real block $J_r^{\mathbb{R}}(\lambda)$. Combining all conjugate pairs and real chains gives a real basis. $\square$

Oral-exam minimum.

- Explain how conjugate complex Jordan chains combine into real blocks.
- Write the block R_{a+ib} .
- Distinguish real Jordan form from complex Jordan form.

Practice problems.

1. Find the real Jordan form of a real matrix with complex eigenvalues $2 \pm 3i$, each with one Jordan block of size 2.
2. Prove that a real matrix of odd size has at least one real eigenvalue.
3. Show how the real block changes if one uses the basis $(y_1, x_1, \dots)$ instead.

13 Normal operators and the spectral theorem

Core idea. Self-adjoint, skew-adjoint, and unitary operators are best understood as special cases of normal operators.

Definition 13.1. Let V be a finite-dimensional complex inner-product space. The adjoint of $A \in \text{End}(V)$ is the unique operator A^* satisfying

$$\langle Av, w \rangle = \langle v, A^*w \rangle \quad (v, w \in V).$$

An operator is normal if

$$A^*A = AA^*.$$

Lemma 13.2. *If A is normal, then*

$$\text{Ker}(A - \lambda I) = \text{Ker}(A^* - \bar{\lambda}I).$$

Proof. For any v , normality of $A - \lambda I$ gives

$$\|(A - \lambda I)v\|^2 = \|(A^* - \bar{\lambda}I)v\|^2.$$

Thus one side is zero if and only if the other is zero. □

Theorem 13.3 (Spectral theorem for complex normal operators). *If A is normal on a finite-dimensional complex inner-product space, then V has an orthonormal basis of eigenvectors of A .*

Proof. Since $\mathbb{C}$ is algebraically closed, A has an eigenvector v . Normalize it. Let $Av = \lambda v$. By the lemma, $A^*v = \bar{\lambda}v$. Therefore the orthogonal complement $v^\perp$ is invariant under A : if $w \perp v$, then

$$\langle Aw, v \rangle = \langle w, A^*v \rangle = \langle w, \bar{\lambda}v \rangle = 0.$$

The restriction of a normal operator to this invariant orthogonal complement is normal. Apply induction on $\dim V$. □

Oral-exam minimum.

- Define adjoint and normal operator.
- Prove the spectral theorem for normal operators.
- Use it as the common source for self-adjoint, skew-adjoint, and unitary cases.

Practice problems.

1. Prove that every normal triangular matrix is diagonal.
2. Show that eigenspaces of a normal operator for distinct eigenvalues are orthogonal.
3. Give an example of a diagonalizable matrix that is not normal with respect to the standard inner product.

14 Self-adjoint operators

Core idea. Self-adjoint operators are precisely the normal operators whose eigenvalues are real; their eigenspaces are orthogonal.

Definition 14.1. An operator A on an inner-product space is self-adjoint if

$$A^* = A.$$

Over $\mathbb{R}$, this is the same as a symmetric matrix in an orthonormal basis. Over $\mathbb{C}$, it is a Hermitian matrix.

Theorem 14.2 (Spectral theorem for self-adjoint operators). *Let $A = A^*$ on a finite-dimensional complex inner-product space. Then there exists an orthonormal basis of V consisting of eigenvectors of A , and all eigenvalues are real. Over $\mathbb{R}$, a real self-adjoint operator has an orthonormal eigenbasis over $\mathbb{R}$.*

Proof. Self-adjoint operators are normal, so over $\mathbb{C}$ the normal spectral theorem gives an orthonormal eigenbasis. If $Av = \lambda v$, $v \neq 0$, then

$$\lambda\|v\|^2 = \langle Av, v \rangle = \langle v, Av \rangle = \bar{\lambda}\|v\|^2,$$

so $\lambda \in \mathbb{R}$. For a real self-adjoint operator, complexification gives real eigenvalues and conjugation-invariant eigenspaces. Each eigenspace has a real orthonormal basis. Combining them gives a real orthonormal eigenbasis. $\square$

Corollary 14.3. *A self-adjoint operator is diagonalizable by a unitary matrix over $\mathbb{C}$ and by an orthogonal matrix over $\mathbb{R}$.*

Oral-exam minimum.

- Define self-adjoint operator.
- Prove eigenvalues are real.
- State and prove the spectral theorem using the normal theorem.

Practice problems.

1. Prove that $\langle Av, v \rangle \in \mathbb{R}$ for self-adjoint A .
2. Diagonalize a real symmetric 2×2 matrix orthogonally.
3. Prove that if $A = A^*$ and $\langle Av, v \rangle = 0$ for all v , then $A = 0$.

15 Skew-adjoint operators

Core idea. Skew-adjoint operators become self-adjoint after multiplication by i . Thus their complex eigenvalues are purely imaginary.

Definition 15.1. An operator A is skew-adjoint if

$$A^* = -A.$$

Over $\mathbb{R}$, this means the matrix is skew-symmetric in an orthonormal basis.

Theorem 15.2 (Spectral theorem for skew-adjoint operators). *Let $A^* = -A$ on a finite-dimensional complex inner-product space. Then V has an orthonormal eigenbasis for A , and every eigenvalue lies in $i\mathbb{R}$. Over $\mathbb{R}$, there is an orthonormal basis in which A is a block diagonal sum of blocks*

$$\begin{pmatrix} 0 & -\mu \\ \mu & 0 \end{pmatrix}$$

and possibly a zero block.

Proof. Over $\mathbb{C}$, the operator $B = -iA$ satisfies

$$B^* = iA^* = i(-A) = -iA = B,$$

so B is self-adjoint. Hence B has an orthonormal eigenbasis with real eigenvalues μ . Then

$$Bv = \mu v \iff Av = i\mu v.$$

Thus eigenvalues of A are purely imaginary. For real A , complex eigenvalues occur as conjugate pairs $\pm i\mu$. The real and imaginary parts of a complex eigenvector give a real invariant plane on which the matrix is the displayed 2×2 block. The kernel gives the zero block. $\square$

Oral-exam minimum.

- Define skew-adjoint.
- Reduce the complex theorem to self-adjointness of $-iA$.
- State the real block form.

Practice problems.

1. Prove that a real skew-symmetric matrix has determinant 0 in odd dimension.
2. Find the real canonical form of $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$.
3. Prove that e^{tA} is unitary/orthogonal if $A^* = -A$.

16 Unitary and orthogonal operators

Core idea. Unitary operators preserve the inner product; their complex eigenvalues lie on the unit circle.

Definition 16.1. An operator U on a complex inner-product space is unitary if

$$U^*U = UU^* = I.$$

On a real inner-product space it is called orthogonal.

Proposition 16.2. *An operator U is unitary if and only if*

$$\langle Uv, Uw \rangle = \langle v, w \rangle$$

for all v, w . Equivalently, it preserves norms.

Proof. If $U^*U = I$, then

$$\langle Uv, Uw \rangle = \langle v, U^*Uw \rangle = \langle v, w \rangle.$$

Conversely, if inner products are preserved, then $\langle (U^*U - I)v, w \rangle = 0$ for all v, w , so $U^*U = I$. In finite dimension this implies U is invertible with $U^{-1} = U^*$, hence $UU^* = I$. Norm preservation implies inner-product preservation by the polarization identity. $\square$

Theorem 16.3 (Spectral theorem for unitary and orthogonal operators). *Over $\mathbb{C}$, every unitary operator has an orthonormal eigenbasis and all eigenvalues satisfy $|\lambda| = 1$. Over $\mathbb{R}$, every orthogonal operator has an orthonormal basis in which its matrix is a block diagonal sum of 1, -1 , and planar rotation blocks*

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Proof. A unitary operator is normal because $U^*U = UU^* = I$. The complex normal spectral theorem gives an orthonormal eigenbasis. If $Uv = \lambda v$, then

$$\|v\| = \|Uv\| = |\lambda|\|v\|,$$

so $|\lambda| = 1$. For real orthogonal operators, complexification gives eigenvalues on the unit circle. Real eigenvalues are ± 1 . Nonreal conjugate pairs $e^{\pm i\theta}$ combine into real rotation blocks by the same realification argument used for real Jordan form. Since unitary operators are diagonalizable over $\mathbb{C}$, no larger Jordan blocks occur. $\square$

Oral-exam minimum.

- Define unitary and orthogonal operators.
- Prove equivalence with inner-product preservation.
- State complex and real spectral forms.

Practice problems.

1. Prove that an orthogonal matrix has determinant ± 1 .
2. Classify all 2×2 real orthogonal matrices.
3. Prove that if U is unitary, then $U^{-1} = U^*$.

17 Bilinear forms

Core idea. A bilinear form is a tensor in $V^* \otimes V^*$; its matrix is a coordinate representation and transforms by congruence, not similarity.

Definition 17.1. A bilinear form on a vector space V over $\mathbb{F}$ is a map

$$B : V \times V \rightarrow \mathbb{F}$$

that is linear in each argument. It is symmetric if

$$B(x, y) = B(y, x),$$

and skew-symmetric if

$$B(x, y) = -B(y, x).$$

Definition 17.2. Given a basis $e = (e_1, \dots, e_n)$, the matrix of B is

$$[B]_e = (b_{ij}), \quad b_{ij} = B(e_i, e_j).$$

If $[x]_e = X$ and $[y]_e = Y$, then

$$B(x, y) = X^T [B]_e Y.$$

Proposition 17.3 (Change of basis). *If $e' = eS$, meaning the columns of S are the coordinates of the new basis vectors in the old basis, then*

$$[B]_{e'} = S^T [B]_e S.$$

Proof. For coordinate columns X', Y' in the new basis, the old coordinates are $X = SX', Y = SY'$. Hence

$$B(x, y) = X^T [B]_e Y = (X')^T S^T [B]_e SY'.$$

Therefore the new matrix is $S^T [B]_e S$. □

Proposition 17.4. *Assume $\text{char } \mathbb{F} \neq 2$. Every bilinear form decomposes uniquely as*

$$B = B_s + B_a,$$

where

$$B_s(x, y) = \frac{B(x, y) + B(y, x)}{2}, \quad B_a(x, y) = \frac{B(x, y) - B(y, x)}{2}$$

are symmetric and skew-symmetric respectively.

Proof. The displayed formulas plainly define symmetric and skew-symmetric bilinear forms, and their sum is B . If $B = S + A$ with S symmetric and A skew-symmetric, then

$$B(y, x) = S(x, y) - A(x, y).$$

Adding and subtracting the two equations gives the formulas above, so the decomposition is unique. □

Oral-exam minimum.

- Define bilinear, symmetric, and skew-symmetric forms.
- Distinguish similarity $S^{-1}AS$ from congruence S^TBS .
- State the symmetric/skew decomposition.

Practice problems.

1. Find the matrix of $B(x, y) = x_1y_2 + x_2y_1$ and determine whether it is symmetric.
2. Prove that an alternating form satisfies $B(x, x) = 0$. When is the converse true?
3. Compute the change-of-basis matrix for a given bilinear form under a specified basis change.

18 Lax–Milgram theorem

Core idea. Lax–Milgram turns a coercive bilinear form into an isomorphism $H \rightarrow H^*$. It is the analytic version of solving a positive definite linear system.

Definition 18.1. Let H be a real Hilbert space. A bilinear form $a : H \times H \rightarrow \mathbb{R}$ is bounded if there exists $M > 0$ such that

$$|a(u, v)| \leq M\|u\|\|v\|.$$

It is coercive if there exists $\alpha > 0$ such that

$$a(u, u) \geq \alpha\|u\|^2 \quad (u \in H).$$

Theorem 18.2 (Hilbert projection theorem). *Let $M \subset H$ be a closed linear subspace and let $x \in H$. Then there exists a unique $m_0 \in M$ such that*

$$\|x - m_0\| = \inf_{m \in M} \|x - m\|.$$

Moreover,

$$x - m_0 \perp M.$$

Proof. Let $d = \inf_{m \in M} \|x - m\|$, and choose a minimizing sequence $(m_n) \subset M$ with $\|x - m_n\| \rightarrow d$. By the parallelogram identity,

$$\|m_n - m_k\|^2 = 2\|x - m_n\|^2 + 2\|x - m_k\|^2 - 4\left\|x - \frac{m_n + m_k}{2}\right\|^2.$$

Since $(m_n + m_k)/2 \in M$, the last norm is at least d . Letting $n, k \rightarrow \infty$ shows $\|m_n - m_k\| \rightarrow 0$. Hence (m_n) is Cauchy. Because H is complete and M is closed, $m_n \rightarrow m_0 \in M$. Continuity of the norm gives $\|x - m_0\| = d$.

For orthogonality, take any $h \in M$ and any $s \in \mathbb{R}$. Since $m_0 + sh \in M$, the function

$$\phi(s) = \|x - m_0 - sh\|^2$$

has a minimum at $s = 0$. Expanding,

$$\phi(s) = \|x - m_0\|^2 - 2s\langle x - m_0, h \rangle + s^2\|h\|^2.$$

Thus $\phi'(0) = 0$, so $\langle x - m_0, h \rangle = 0$. Therefore $x - m_0 \perp M$. If m_1 is another minimizer, then both $x - m_0$ and $x - m_1$ are orthogonal to M , so $m_1 - m_0 \in M$ is orthogonal to itself; hence $m_1 = m_0$. $\square$

Theorem 18.3 (Riesz representation theorem). *For every bounded linear functional $\ell \in H^*$, there exists a unique $y \in H$ such that*

$$\ell(v) = \langle v, y \rangle \quad (v \in H).$$

Moreover, $\|\ell\| = \|y\|$.

Proof. If $\ell = 0$, take $y = 0$. Otherwise $N = \text{Ker } \ell$ is a closed proper subspace. Choose $x_0 \notin N$. By the Hilbert projection theorem, there is a unique decomposition

$$x_0 = n_0 + z, \quad n_0 \in N, \quad z \in N^\perp, \quad z \neq 0.$$

Then $N^\perp = \text{span}\{z\}$; otherwise two independent vectors in $N^\perp$ would allow a nonzero combination lying in $\text{Ker } \ell = N$, contradiction. For any $v \in H$, the vector

$$v - \frac{\ell(v)}{\ell(z)}z$$

lies in N , hence is orthogonal to z . Thus

$$\langle v, z \rangle = \frac{\ell(v)}{\ell(z)}\|z\|^2,$$

so

$$\ell(v) = \left\langle v, \frac{\ell(z)}{\|z\|^2}z \right\rangle.$$

This gives existence. Uniqueness follows because if $\langle v, y_1 - y_2 \rangle = 0$ for all v , choose $v = y_1 - y_2$. The norm identity follows from Cauchy–Schwarz and equality at $v = y$. $\square$

Theorem 18.4 (Lax–Milgram). *Let H be a real Hilbert space and let $a : H \times H \rightarrow \mathbb{R}$ be bounded and coercive. For every $\ell \in H^*$, there exists a unique $u \in H$ such that*

$$a(u, v) = \ell(v) \quad \forall v \in H.$$

Moreover,

$$\|u\| \leq \frac{1}{\alpha}\|\ell\|.$$

Proof. For each $u \in H$, the map $v \mapsto a(u, v)$ is a bounded linear functional. By Riesz, there is a unique vector $Tu \in H$ such that

$$a(u, v) = \langle Tu, v \rangle \quad (v \in H).$$

The map $T : H \rightarrow H$ is linear and bounded, with $\|T\| \leq M$. Coercivity gives

$$\alpha\|u\|^2 \leq a(u, u) = \langle Tu, u \rangle \leq \|Tu\|\|u\|,$$

so

$$\|Tu\| \geq \alpha\|u\|.$$

Hence T is injective and $\text{Im } T$ is closed: if $Tu_n \rightarrow y$, then u_n is Cauchy because $\|u_n - u_m\| \leq \alpha^{-1}\|Tu_n - Tu_m\|$, so $u_n \rightarrow u$ and $Tu = y$. The range is dense: if $w \perp \text{Im } T$, then $\langle Tu, w \rangle = 0$ for all u , so $a(u, w) = 0$ for all u . Taking $u = w$ gives $a(w, w) = 0$, hence $w = 0$. Thus $\text{Im } T$ is both closed and dense, so $\text{Im } T = H$.

By Riesz, choose $y \in H$ such that $\ell(v) = \langle y, v \rangle$. Since T is onto, choose u with $Tu = y$. Then

$$a(u, v) = \langle Tu, v \rangle = \ell(v).$$

Uniqueness follows from coercivity: if $a(u_1 - u_2, v) = 0$ for all v , take $v = u_1 - u_2$. Finally,

$$\alpha\|u\|^2 \leq a(u, u) = \ell(u) \leq \|\ell\|\|u\|,$$

so $\|u\| \leq \alpha^{-1}\|\ell\|$. $\square$

Oral-exam minimum.

- Define boundedness and coercivity.
- State Riesz representation and Lax–Milgram.
- Prove Lax–Milgram through the operator T .

Practice problems.

1. Prove Lax–Milgram in finite dimension using matrices.
2. Give an example of a bounded bilinear form that is not coercive.
3. Let $a(u, v) = \int_0^1 u'v' + uv \, dx$ on $H_0^1(0, 1)$. Identify why it is coercive.

19 Quadratic forms and change of basis

Core idea. A quadratic form is the diagonal trace of a symmetric bilinear form; its matrix also transforms by congruence.

Definition 19.1. A quadratic form on V over a field of characteristic not 2 is a function

$$Q : V \rightarrow \mathbb{F}$$

of the form

$$Q(x) = B(x, x)$$

for a symmetric bilinear form B . The associated symmetric bilinear form is recovered by polarization:

$$B(x, y) = \frac{1}{2}(Q(x + y) - Q(x) - Q(y)).$$

Proof of polarization. If $Q(x) = B(x, x)$ and B is symmetric, then

$$Q(x + y) = B(x + y, x + y) = B(x, x) + 2B(x, y) + B(y, y).$$

Rearranging gives the formula. □

Definition 19.2. In a basis e , the matrix of Q is the symmetric matrix S satisfying

$$Q(x) = X^T S X, \quad X = [x]_e.$$

Proposition 19.3 (Change of basis). *If $e' = eC$, then*

$$[Q]_{e'} = C^T [Q]_e C.$$

Proof. This is the bilinear-form congruence formula applied to the polar form of Q . Equivalently, if $X = CX'$, then

$$Q(x) = X^T S X = (X')^T C^T S C X'.$$

□

Remark 19.4. Operator matrices transform by similarity $C^{-1}AC$. Form matrices transform by congruence $C^T SC$. This distinction is essential.

Oral-exam minimum.

- Define quadratic form and polar bilinear form.
- Prove the polarization formula.
- State the congruence transformation law.

Practice problems.

1. Find the matrix of $Q(x, y, z) = x^2 + 2xy + 3z^2$.
2. Compute the polar form of $Q(x, y) = x^2 - xy + 4y^2$.
3. Show that different symmetric matrices cannot define the same quadratic form if $\text{char } \mathbb{F} \neq 2$.

20 Diagonalization and Sylvester's law of inertia

Core idea. Symmetric quadratic forms are classified over $\mathbb{R}$ by the numbers of positive, negative, and zero squares.

Theorem 20.1 (Diagonalization of quadratic forms). *Let $\text{char } \mathbb{F} \neq 2$. Every quadratic form on a finite-dimensional vector space has a diagonal matrix in some basis.*

Proof. Use induction on $n = \dim V$. If $Q = 0$, any basis works. If there exists v with $Q(v) \neq 0$, let B be the polar form. Then

$$V = \mathbb{F}v \oplus v^\perp, \quad v^\perp = \{w : B(v, w) = 0\}.$$

Indeed, for any x , set

$$c = \frac{B(x, v)}{B(v, v)}$$

and write $x = cv + (x - cv)$, with $x - cv \in v^\perp$. Apply induction to $Q|_{v^\perp}$. If $Q(v) = 0$ for all v but $B \neq 0$, then choose x, y with $B(x, y) \neq 0$. Since $Q(x + y) = 2B(x, y) \neq 0$, this reduces to the previous case. $\square$

Theorem 20.2 (Sylvester's law of inertia). *Over $\mathbb{R}$, every quadratic form has a diagonal representation*

$$Q = x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_{p+q}^2,$$

possibly with additional zero variables. The triple

$$(p, q, r), \quad r = \text{nullity } Q,$$

is independent of the diagonalizing basis.

Proof. Diagonalization gives coefficients a_i . Rescaling basis vectors changes nonzero a_i to ± 1 , so such a form exists. Let P be the maximal dimension of a subspace on which Q is positive definite. In the displayed normal form, the span of the first p basis vectors is positive definite, so $P \geq p$. Any subspace of dimension $p + 1$ must intersect the subspace spanned by the negative and zero variables nontrivially, by dimension counting; on that intersection $Q \leq 0$, so the subspace cannot be positive definite. Hence $P = p$. Thus p is invariant. Similarly, q is the maximal dimension of a negative definite subspace. The nullity r is the dimension of the radical

$$\text{Rad}(B) = \{v : B(v, w) = 0 \ \forall w\},$$

which is basis-independent. Therefore the inertia triple is invariant. $\square$

Oral-exam minimum.

- Prove diagonalization by orthogonal splitting relative to the polar form.
- State Sylvester's law of inertia.
- Explain why congruence, not similarity, is the equivalence relation.

Practice problems.

1. Diagonalize $Q(x, y) = 2x^2 + 4xy + 3y^2$.
2. Find the inertia of $Q(x, y, z) = x^2 + y^2 - z^2 + 2xy$.
3. Prove that a real symmetric matrix is positive definite iff all eigenvalues are positive.

21 Rayleigh quotient and stationary eigenvectors

Core idea. For a self-adjoint operator, eigenvectors are precisely stationary points of the Rayleigh quotient on the projective space of lines.

Definition 21.1. Let $A = A^*$ on a finite-dimensional real or complex inner-product space. The Rayleigh quotient is

$$R_A(x) = \frac{\langle Ax, x \rangle}{\langle x, x \rangle}, \quad x \neq 0.$$

It is constant on nonzero scalar multiples of x , so it is naturally a function on lines.

Theorem 21.2 (Stationary points of the Rayleigh quotient). *Let $A = A^*$. A nonzero vector x is a stationary point of R_A if and only if x is an eigenvector of A .*

Proof. It is enough to restrict to the unit sphere $S = \{x : \|x\| = 1\}$. On S ,

$$R_A(x) = \langle Ax, x \rangle.$$

Let h be tangent to the sphere at x , so $\Re\langle h, x \rangle = 0$. The first variation is

$$dR_A(x)[h] = \langle Ah, x \rangle + \langle Ax, h \rangle = 2\Re\langle Ax, h \rangle.$$

If x is stationary, then this is zero for every tangent h . Thus Ax is orthogonal to every tangent vector. The orthogonal complement of the tangent space at x is $\text{span}\{x\}$, hence $Ax = \lambda x$. Since A is self-adjoint, $\lambda = R_A(x) \in \mathbb{R}$. Conversely, if $Ax = \lambda x$, then for tangent h ,

$$dR_A(x)[h] = 2\Re\langle \lambda x, h \rangle = 0,$$

so x is stationary. □

Theorem 21.3 (Min-max endpoints). *If $A = A^*$ has eigenvalues*

$$\lambda_1 \leq \cdots \leq \lambda_n,$$

then

$$\lambda_1 = \min_{x \neq 0} R_A(x), \quad \lambda_n = \max_{x \neq 0} R_A(x).$$

Proof. Choose an orthonormal eigenbasis e_i . If $x = \sum c_i e_i$, then

$$R_A(x) = \frac{\sum_i \lambda_i |c_i|^2}{\sum_i |c_i|^2}.$$

This is a weighted average of the eigenvalues, so it lies between λ_1 and λ_n . Equality is achieved at eigenvectors for λ_1 and λ_n . □

Oral-exam minimum.

- Define the Rayleigh quotient.
- Prove stationary point iff eigenvector.
- Derive the min and max eigenvalue characterization.

Practice problems.

1. Compute the Rayleigh quotient of a real symmetric 2×2 matrix and find its stationary points.
2. Prove that $R_A(x)$ is real for self-adjoint A .
3. Show that if A is positive definite, then $R_A(x) > 0$ for all $x \neq 0$.

Final exam checklist

Before the exam, verify that you can do the following without notes:

1. Treat V with an operator A as an $\mathbb{F}[t]$ -module.
2. Prove the cyclic module lemma and build companion matrices.
3. State and prove the role of annihilating and minimal polynomials.
4. Derive Frobenius normal form from invariant factors.
5. Prove Cayley–Hamilton using Frobenius form.
6. Prove primary decomposition and construct Jordan bases.
7. Compute polynomials and functions of Jordan blocks.
8. Explain Danilevsky’s method as similarity to companion form.
9. Construct eigenvectors and generalized eigenvectors of companion matrices.
10. Complexify a real vector space and operator.
11. Build real Jordan blocks from conjugate complex chains.
12. Prove the spectral theorem for normal operators and its self-adjoint, skew-adjoint, and unitary consequences.
13. Work with bilinear forms as coordinate-free tensors and compute congruence transformations.
14. State and prove Lax–Milgram through the Riesz map.
15. Diagonalize quadratic forms and prove Sylvester’s law of inertia.
16. Prove the Rayleigh quotient stationary point theorem.